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Treadmill-based Gait Analysis using Camera Kinematic System, Force Sensing Resistor (FSR) and Motion Processing Unit (MPU) Sensors

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Abstract

The systematic study of human gait patterns is generally done through normal observation in Nepal which is not quite reliable. Our project aims at designing and executing instrumented and systemic gait analysis system using camera kinematic system, MPU and FSR sensors to study various gait-related parameters of the experimental subjects walking on the treadmill. The preliminary stage involves calibration and testing of hardware components including MPU and FSR sensors along with image processing system. Video processing system employs image processing techniques for each frame in the video in order to detect joint angles of the lower extremities. Likewise, six MPU sensors via multiplexer and eight FSR sensors were connected to Arduino Mega programmed using Arduino IDE. MPU sensors provide acceleration data whereas FSR sensors calculate force exertion at different regions of the foot. These data from Arduino were transmitted wirelessly via Bluetooth module to an Android app in the mobile phone. Data collection of 15 normal experimental subjects along with 3 knee osteoarthritis patients was conducted and data analysis was finally performed in MS-Excel to infer relevant information about gait-related parameters. Normal subjects exhibited joint angles within the standard range of the previous studies while abnormal subjects significantly showed reduced knee flexion. Almost similar nature of amplitude variation of force exertion was observed among different regions of the foot in normal subjects while force distribution was notably concentrated in the right foot of abnormal subjects due to the problem in their left leg. The nature of acceleration graphs appeared basically similar among these two study groups.

Keywords: *Gait analysis, calibration, video processing system, image processing, data collection and data analysis.*

Abbreviations

MPU	Motion Processing Unit
FSR	Force Sensing Resistor
CT	Computed Tomography
MRI	Magnetic Resonance Imaging
EMG	Electromyography
KEMG	Kinesiological Electromyography
sEMG	Surface Electromyography
AI	Artificial Intelligence
CCD	Charge-Coupled Device
CNN	Convolutional Neural Network
GRF	Ground Reaction Force
IMU	Inertial Measurement System
MS	Multiple Sclerosis
BPPV	Benign Paroxysmal Positional Vertigo
MEMS	Micro Electro-mechanical System
USB	Universal Serial Bus
ICSP	In- circuit Serial Programming
LED	Light Emitting Diode
I2C	Inter-Integrated Circuit
IR	Infrared Radiation
CSB	Canadian Society of Biomechanics

Table of Contents

Acknowledgement ..... **i**

Abstract ..... **ii**

Abbreviations ..... **iii**

Table of Contents ..... **iv**

List of Figures ..... **vi**

List of Tables ..... **viii**

Chapter 1 Introduction **1**

1.1 Background 1

1.2 Problem Statement 3

1.3 Objectives 4

1.3.1 General Objective 4

1.3.2 Specific Objectives 4

Chapter 2 Literature Review **5**

2.1 Overview of gait analysis 5

2.1.1 Phases of a gait 6

2.1.2 History of gait analysis 7

2.1.3 Applications of gait analysis 9

2.2 Biomechanical factors influencing gait 11

2.3 General factors affecting gait 12

2.4 Techniques used for gait analysis 13

2.4.1 Use of treadmill in gait analysis 14

2.5 Gait abnormalities and related disorders 16

2.5.1 Gait analysis for knee osteoarthritis 18

Chapter 3 Materials and Methodology **19**

3.1 Materials 19

3.1.1 MPU sensor 19

3.1.2 FSR sensor 19
3.1.3 Arduino Mega 20
3.1.4 Multiplexer 20
3.1.5 Bluetooth module 20
3.1.6 Active markers 21
3.1.7 Velcro 21
3.1.8 Ribbon cable 21
3.1.9 Shoe sole 22
3.1.10 Treadmill 22
3.2 Methodology 23
3.2.1 Overview of the system 23
3.2.2 Data collection 32
3.2.3 Data analysis 32
Chapter 4 Results 34
4.1 Video processing system 34
4.2 Wearable sensors 37
4.3 Output of the sensors 38
4.3.1 FSR sensors 38
4.3.2 MPU sensors 40
Chapter 5 Discussion 42
Chapter 6 Conclusion and Further Works 45
Total Cost 47
Gantt Chart 48
References 49
Appendices 58

List of Figures

Figure 2.1 Events in a gait cycle 6
Figure 2.2 Patterns of various gait disorders 18
Figure 3.1 MPU 6050 19
Figure 3.2 FSR UX 402 19
Figure 3.3 Arduino Mega 2560 20
Figure 3.4 MUX TCA9548A 20
Figure 3.5 Bluetooth module HC-05 21
Figure 3.6 Marker (X-base: 14mm) 21
Figure 3.7 Velcro 21
Figure 3.8 Ribbon cable 22
Figure 3.9 Shoe sole 22
Figure 3.10 Treadmill WNQ model 22
Figure 3.11 Block diagram showing overall methodology of the gait analysis system 23
Figure 3.12 Image processing steps involved in each video frame during video processing 25
Figure 3.13 Representation of segment angles as per the CSB gait standard 26
Figure 3.14 Flowchart showing the steps involved in video processing 27
Figure 3.15 Plot of mass vs. conductance with a fitted straight line 28
Figure 3.16 Block diagram showing MPU 6050 and FSR interfacing to Arduino Mega 2560 29
Figure 3.17 Circuit diagram involving MPU sensors for (a) transmitter side and (b) receiver side 30
Figure 3.18 Circuit diagram involving FSR sensors for (a) transmitter side and (b) receiver side 31
Figure 4.1 Graphical representation of joint angles during one complete gait cycle for (a) standard reference (b) normal subject group and (c) abnormal subject group 36
Figure 4.2 Development of (a) MPU interfaced system and (b) FSR interfaced system 37
Figure 4.3 Force distribution of right and left toes for (a) normal and (b) abnormal subject groups 38
Figure 4.4 Force distribution of right and left first metatarsals for (a) normal and (b) abnormal subject groups 38

Figure 4.5 Force distribution of right and left fifth metatarsals for (a) normal and (b) abnormal subject groups 39
Figure 4.6 Force distribution of right and left heel regions for (a) normal and (b) abnormal subject groups 39
Figure 4.7 Nature of motion of right and left thigh regions for (a) normal and (b) abnormal subject groups 40
Figure 4.8 Nature of motion of right and left shank regions for (a) normal and (b) abnormal subject groups 40
Figure 4.9 Nature of motion of right and left foot regions for (a) normal and (b) abnormal subject groups 41

List of Tables

Table 2.1 Gait parameters and their description 11
Table 2.2 Phenomenological classification of gait disorders.....16
Table 3.1 Comparison of joint angles obtained manually and through image processing system.....24
Table 4.1 Comparison of joint angles between the normal and abnormal subject groups during different gait phases.....34

Chapter 1 Introduction

1.1 Background

Gait provides significant information about the specific movements and walking patterns of an individual. It can vary from person to person and can also be affected by factors such as age, health conditions and injuries. Gait abnormality may indicate any underlying issues or potential health problems, making gait analysis a valuable tool for diagnosis, treatment planning and monitoring the progress of certain clinical conditions. Gait deviations can be indicative of various health conditions such as neurological disorders, musculoskeletal issues, cardiovascular problems and even cognitive impairments. Neurodegenerative disorders such as Parkinson's disease, multiple sclerosis and certain types of dementia can often be predicted through specific changes in gait patterns. Tremors, shuffling steps, reduced arm swing and balance problems are some examples of gait changes that can help in diagnosing these diseases [1]. Gait analysis can also aid in developing effective treatment planning and monitoring the patient's improvement over time which is useful in rehabilitation programs allowing healthcare providers to adjust therapies and interventions.

Gait analysis involves gait parameters such as stride length, step length, cadence, walking speed, joint angles and ground reaction forces by employing techniques such as motion capture cameras, force plates, EMG, pressure mats, MPU and video analysis methods to collect data for analysis and identify potential abnormalities or issues in the walking pattern [2]. Assessment of joint angles such as hip, knee and ankle angles during the gait cycle provides insights into joint movement patterns, alignment and potential issues such as joint stiffness or muscle weakness. Force plates and pressure mats are used to determine force exerted by the person's feet on the ground during walking. These force offer information about weight distribution, balance and how different parts of the body interact with the ground. They reveal how weight is shifted between different regions of the feet and help diagnose foot-related abnormalities. Similarly, MPU sensor measures the orientation and motion of body segments.

In Nepal, there is currently no any standard technique used for analyzing and evaluating the gait patterns of the individual. Medical professionals assess and analyze the walking patterns of the subject solely through visual observation, which may not always be reliable and authentic. Only limited number of studies and research projects have been conducted in the field of gait analysis, which has hindered the ability of medical professionals to fully

understand and address gait abnormalities and related health issues to some extent. On the other hand, the developed nations with sufficient resources and advanced facilities employ sophisticated cameras, sensors and technologies to conduct gait analysis efficiently and more systematically. Additionally, the methods used in analyzing the gait parameters are either too primitive or way too complex, expensive and requires expertise. So, designing a system that is simple yet affordable can help medical personnel in diagnosing gait abnormalities and their impact on the individual's health.

Hence, our project can serve as a stepping stone for the advancement of gait analysis system in Nepal. We focus on developing a simple and inexpensive method for conducting gait analysis by incorporating the use of treadmill, Motion Processing Unit (MPU) sensors, Force Sensing Resistor (FSR) sensors as well as camera kinematic system. This integrated approach aims to provide comprehensive and effective solution for studying an individual's gait pattern. Our goal is to create an in-shoe system equipped with the FSR sensors to obtain the measure of force exerted in different locations of the foot. In addition, the MPU sensors have been incorporated which are placed on the thigh, shank and foot regions of the lower extremities to record the acceleration values during different gait phases. Furthermore, in our project, a stationary camera is positioned parallel to the right sagittal plane of the experimental subject with active markers attached to the different joint locations. This camera captures the video of these experimental subjects walking on the treadmill and helps to compute the joint angles through image processing techniques. For the purpose of image processing, Python has been used and MS-Excel has been employed for further processing and analysis of the collected data.

Implication of our project lies in identifying and visualizing the normal gait patterns as well as abnormal gait patterns associated with different clinical conditions. It concentrates on collecting data of gait parameters in normal subjects and abnormal patients with particular emphasis on the conditions of knee osteoarthritis followed by plotting graphs of those parameters in the experimental subjects. Through graph plots, the comparison of the parameters was drawn between the normal and abnormal subjects and the potential differences or variations were noted in both cases. The data collected in our project can serve as a reference for further research and advancement in the field of gait analysis. Our project has utilized readily accessible sensors and a simplified system design, rendering it both cost effective and user friendly. Thus, the overall project on gait analysis is a complete package of system design, software implementation, video processing and statistical analysis that requires knowledge from the multidisciplinary subjects.

1.2 Problem Statement

Various gait analysis techniques have been developed so far which are commercially used in many countries. However, there are several challenges related to gait analysis in Nepal that led to initiation of our project which are outlined below:

a. Traditional gait analysis methods are more time consuming but gait analysis over treadmill requires relatively less time for data collection.
b. Pre-existing techniques requires specific gait lab for gait analysis but with the help of wearable sensors, gait analysis can be performed conveniently wherever required.
c. The expensive nature of gait analysis system has created barriers to its accessibility in the healthcare sector. Hence, a cheaper alternative may be proposed so as to analyze the different gait parameters and aid in rehabilitation of patients suffering from gait related disorders.
d. Use of many cameras to capture the image and video of the walking pattern is more prevalent in current scenario which increases the complexity of processing. So, our aim is to perform gait analysis using minimum number of cameras that can make the system less complex and inexpensive.
e. Gait analysis using embedded force plates requires the subject to step-on the fixated specific positions. So, the use of FSR sensors on the in-sole system helps to fix such problem.
f. In Nepal there is no provision for collection, visualization and analysis of data during gait analysis which could be utilized for the future reference. Hence, data collection done during our project could be of significant use for future research and advancement.
g. Despite continuous study of human motion over several years, clinical aspects of gait interpretation has not yet progressed. It is, therefore, necessary to provide a convenient system capable of providing proper data collection methods and relevant data analysis techniques in gait analysis.

1.3 Objectives

The objectives of the project are as follows:

1.3.1 General Objective

To perform treadmill-based gait analysis using camera kinematic system, FSR and MPU sensors.

1.3.2 Specific Objectives

a. To design an in-sole system using FSR sensors to measure the applied force on the foot and employ MPU sensors to observe the nature of motion in lower extremities of the body.

b. To develop marker-based system and capture video of the moving subject on the treadmill using stationary camera.

c. To use image processing techniques to determine the position of these markers and carry out appropriate procedure for joint angle measurement.

d. To collect data and utilize them for the processing and analysis to derive some relevant findings.

e. To compare the gait-related parameters between normal healthy subjects and gait abnormality patients with knee osteoarthritis.

Chapter 2 Literature Review

2.1 Overview of gait analysis

Gait analysis is a special branch of biomechanics that provides objective examination of walking patterns with the aim of revealing the key links and influencing factors of gait abnormality, so as to guide rehabilitation assessment and treatment [3]. It deals with the technical aspects of the gait assessment including detailed visual examination of the individual's gait and making objective measurements of the parameters such as joint angles, ground reaction force, cadence, stride length, speed and others [2].

A normal gait refers to a series of rhythmic, alternating movements of the lower extremities that results in the forward progression of the body [4]. Primarily, gait analysis focuses on detecting any abnormality in the human walking pattern [5]. It precisely measures the movement of the body parts, body mechanics and muscle activities providing the dynamic status of the body that are not observed through the reports of the X-rays, CT scan or MRI.

Gait analysis is applicable mainly in the determination of gait functionalities, irregularities and classification of gait abnormalities so that better remedies can be explored for non-ambulatory patients and gait enhancement facilities can be utilized for military or disaster recovery applications [6]. Hence, it offers assistive solutions in almost all considerable fields of human locomotion including rehabilitation medicine, orthopedics, kinesiology, sports science and other related fields for both healthy as well as pathological subjects [7].

Modern technology, including wearable sensors, force plates, and motion capture devices, is increasingly used in gait analysis. With the aid of these technologies, researchers and medical professionals are able to collect precise data for an in-depth assessment [8]. Gait analysis is essential for customizing rehabilitation treatments along with the assessment. It aids therapists in keeping track of patients' recovery following accidents or surgeries, making necessary adjustments to interventions and setting reasonable goals [9].

Gait analysis is also used in the field of human-computer interaction to create prosthetics, exoskeletons and aids that adapt to each individual's particular gait patterns [10]. It extends further addressing already-present complications. By enhancing their gait patterns and locating possible risk factors, it can also be used in sports science and biomechanics to prevent injuries [11]. Gait analysis can support the early detection of mobility difficulties

and guide interventions to enhance the quality of life for elderly people [12]. Furthermore, clinical practice is greatly influenced by gait analysis. For those with musculoskeletal or neurological disorders, it helps to identify certain gait irregularities, enabling accurate diagnosis and customized treatment methods [13].

2.1.1 Phases of a gait

Human gait involves intricate co-ordination between the head, trunk, arms and lower limbs with the primary objective of getting from one place to another safely and effectively. The time of foot contact with the ground at different instances can better explain these cyclic interactions [14]. A gait cycle is a comprehensive series of the actions performed by the limbs as illustrated in the Figure 2.1.

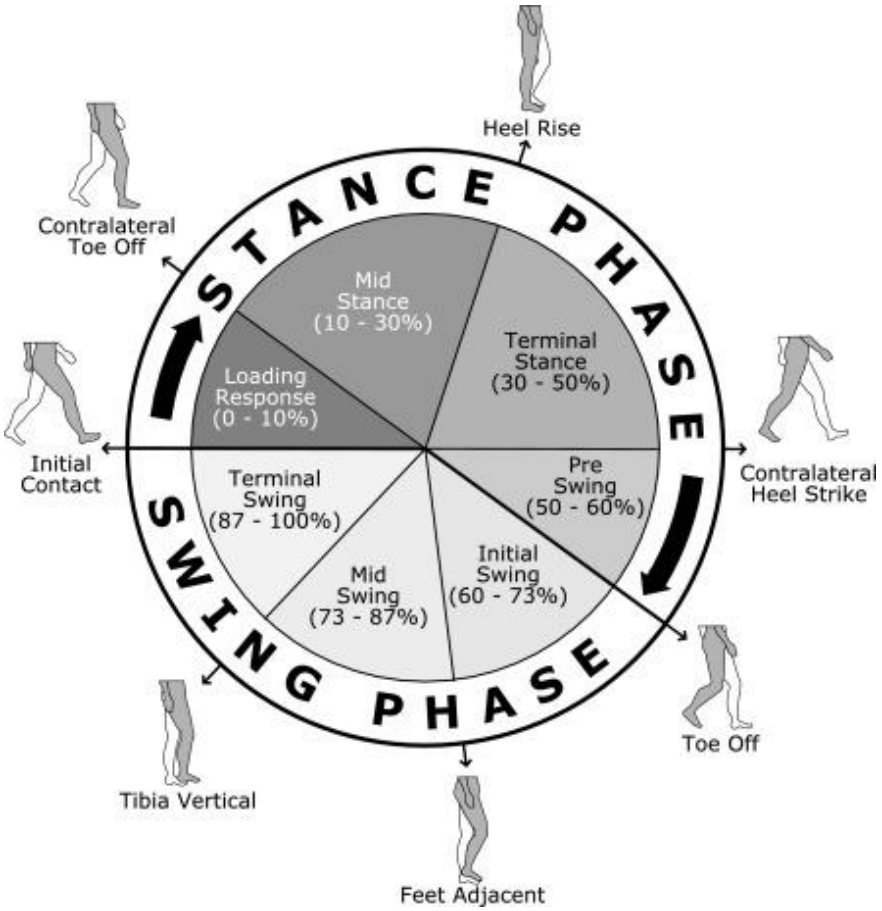


Figure 2.1 Events in a gait cycle [15]

Each gait cycle comprises of two main phases, namely stance and swing. The time during which the foot is in contact with the ground marks the stance phase. Likewise, the swing period, which comes after the stance period, is indicated by the elevation of the same foot above the ground [15].

The stance phase can be divided into four sub-phases: loading response, mid-stance, terminal stance and pre swing. Loading response begins when the foot initially touches the ground through the heel region with a supinated position. During mid-stance, the contralateral foot is off the ground requiring the reference foot to bear the whole body weight alone [16]. Terminal stance begins when the heel region of the foot is raised off the ground and ends when the heel region of the opposite foot finally strikes the ground. Pre-swing is the final part of stance phase where the contralateral foot rests in supine position and the toe of the foot finally leaves the ground.

Similarly, the swing phase can be divided into three sub-phases: initial swing, mid-swing and terminal-swing. During initial swing, the foot is lifted and advanced by hip flexion, increased knee flexion and partial dorsiflexion of the ankle with the other foot in its early mid-stance [17]. Mid-swing takes place as the swinging foot goes beside the stance foot and the tibia lies vertical to the ground. Finally, terminal swing is the deceleration phase in which the extremity is decelerating in an attempt for heel strike and ends just prior to the initial foot contact [16].

2.1.2 History of gait analysis

Aristotle (384–322 BCE) can be recognized for providing the preliminary concept of gait who made the first written comments on the patterns of human walking but could not hold them true considering difficulty to perform experimental tests of the propositions at that time. Ancient civilizations, such as the Greeks and Romans, made some early observations about walking and running, but these were often based on anecdotal accounts rather than systematic analysis. The advent of gait analysis officially dates back to the 17th century in Europe [18] when mathematical basis for modern gait analysis came to spotlight attributed to the Renaissance era [19]. During the Renaissance period, anatomists like Leonardo da Vinci began to study human anatomy and made detailed drawings of the musculoskeletal system which laid the foundation for a better understanding of human movement.

The scientific foundation in progress of gait analysis study can be accredited to Borelli, Galvani, Newton, Descartes, Marey, Carlet, Weber brothers, Scherb, Duchenne, Muybridge, and Braune and Fischer [18]. The 19th century saw significant advancements in the understanding of biomechanics and locomotion. Eadweard Muybridge, an English photographer, conducted a series of motion-capture experiments using multiple cameras to capture the movements of humans and animals [20]. His work provided valuable insights into the mechanics of gait. In the early 20th century, the development of cinematography

allowed for more precise analysis of gait. Researchers began to use high-speed cameras to capture and analyze human movement in greater detail. The study of gait gained importance during the World War II due to the increased need for prosthetic limbs for injured soldiers. This led to the development of tools and techniques to analyze and design prosthetics that could replicate natural walking patterns [21].

Dramatic advancement of gait analysis occurred in the mid-20th century when Vern Inman and colleagues introduced kinesiological electromyography (KEMG), 3-D force and energy measurements in the study of walking in normal subjects and amputees [18]. The profound studies were done and technological advances were stressed enough in an attempt to continue the works of Inman. Furthermore, two surgeons, Jacquelin Perry and David Sutherland contributed to the development of clinical gait analysis. Perry provided methodological approaches to observational gait analysis along with the electromyography (EMG) [22] to apply instrumented methods for the measurement of temporo-spatial parameters [23]. On the other hand, Sutherland advanced methods for the measurement of joint angles with the application of cameras [24].

In Europe, the surgeon Richard Scherb actually pioneered a treadmill-based technique for identifying the phasic activity of different muscles before the advent of reliable EMG [18]. The final development in gait analysis in the pre-computer era was the use of electro-goniometers set forth by engineer Larry Lamoreux so as to determine 3D-joint kinematics [25]. With the dawn of computer era towards the end of the 20th century, the development of modern gait analysis progressively opted for faster processing of reliable data that could be of clinical significance. Researchers began to use motion-capture systems, force plates and sophisticated software to gather and analyze gait data more accurately [19]. This allowed for a better understanding of gait abnormalities and their implications for clinical and research purposes.

Gait analysis became an essential tool in clinical settings, especially in orthopedics, neurology, and rehabilitation. Nowadays, it is used to diagnose and treat various medical conditions such as cerebral palsy, stroke and musculoskeletal disorders [26]. In the realm of sports science, gait analysis can be employed to enhance athletic performance and prevent injuries [27]. Athletes use gait analysis to optimize running or walking techniques and choose appropriate footwear. Today, gait analysis continues to advance with the integration of modern technologies, including 3D motion capture systems, wearable sensors and Artificial Intelligence (AI). These innovations have made gait analysis more accessible and precise [28].

2.1.3 Applications of gait analysis

Gait analysis is a powerful tool in assessing gait with wide range of applications which can be discussed as follows:

a. Sports

In order to maximize performance and minimize injuries, gait analysis is frequently utilized in sports science and athletic training. Experts can spot biomechanical inefficiencies, improper movement patterns and potential improvement areas by examining an athlete's gait [29]. The creation of individualized training plans, suggestions for footwear and corrective workouts can all be done using such information.

b. Clinical diagnosis

Gait analysis provides valuable information for diagnosing musculoskeletal disorders, neurological conditions and other various medical issues. By analyzing an individual's gait patterns, healthcare professionals can identify any abnormalities, deviations or asymmetries associated with the human walking that may indicate underlying conditions [28].

c. Treatment planning

Proper treatment strategies for the patients can be planned with the help of gait analysis. Clinicians can spot specific movement impairments or deficits by measuring gait parameters such step length, stride width, cadence and joint angles [30]. As per this data, various physical therapy methods, assistive equipments and rehabilitation programs can be developed that are specifically targeted for the patients.

d. Rehabilitation

Clinical professionals can objectively evaluate the efficacy of rehabilitation therapy by through gait analysis. Clinicians can monitor improvements in the patient's walking patterns over time by periodically assessing their gait parameters [26]. By modifying treatment strategies and evaluating the results of interventions with the use of this objective feedback, rehabilitation efforts are made to be effective and modified as necessary.

e. Post-surgical assessment

The results of surgical procedure, especially those involving joint replacements or orthopedic surgeries, can be evaluated with the help of gait analysis. Clinicians can evaluate

the outcome of surgery, spot any persistent problems or complications and modify the rehabilitation procedure by comparing pre- and post-operative gait metrics [31].

f. Assistive device customization

Gait analysis is crucial in customizing assistive devices such as prosthetics, orthotics and mobility aids. With the evaluation of the subject's gait patterns, clinicians can optimize the fit, alignment and functionality of these devices to closely mimic natural movement, improve mobility and enhance overall comfort and quality of life [32].

g. Clinical research

Gait analysis is used in clinical research to study the effects of medications, therapies and interventions on gait patterns, especially in drug trials and studies related to neurological disorders [26]. It plays a multifaceted role in clinical research, contributing to the assessment, diagnosis, treatment planning and monitoring of various medical conditions across different medical specialties. Its ability to provide objective, quantifiable data makes it an indispensable tool in improving patient care and advancing medical knowledge.

h. Biomechanical research

Researchers use gait analysis to investigate the mechanics of human movement, leading to a better understanding of how the musculoskeletal system functions. This research can inform the development of new treatments and technologies [33]. It provides quantitative data on joint angles, forces and muscle activity, enabling researchers to understand the mechanics of walking, running and other activities. Gait analysis plays a pivotal role in clinical applications, injury prevention and optimizing sports performance by identifying biomechanical abnormalities, injury risk factors and areas for improvement.

i. Geriatrics

Gait analysis serves as a vital tool for assessing and managing mobility-related issues in elderly individuals [12]. It aids in predicting the risk of falls by analyzing gait patterns and identifying abnormalities, enabling targeted interventions to prevent fall-related injuries. Gait analysis also plays a role in monitoring the progression of age-related conditions such as cognitive decline or dementia, affecting walking ability and balance, facilitating early detection and timely interventions, ultimately contributing to improved quality of life and independence in older adults.

2.2 Biomechanical factors influencing gait

The conventional divisions of gait analysis parameters include: time-distance or spatio-temporal parameters, kinematics parameters, kinetics parameters and EMG parameters (electrical activities of the lower limb muscles) [3] as enlisted in the Table 2.1.

Table 2.1 Gait parameters and their description

Divisions	Gait Parameters	Description	Reference
Spatio-temporal parameters	Step length	Distance from heel of one footprint to heel of opposite footprint.	[34]
	Stride length	Distance between heels of two consecutive footprints of the same foot.	
	Step width	Lateral distance between heels of two consecutive feet.	
	Cadence	Number of foot strikes per minute.	
Kinematics parameters	Body segment positions	Indication of the center of mass of the body.	[35]
	Joint angle	Angle between two segments on either side of the joint.	
	Joint velocity	Angular velocity of one segment relative to another.	
	Joint acceleration	Angular acceleration of one segment relative to another.	
Kinetics parameters	Ground reaction force	Force exerted by ground on the body during each gait cycle.	[36]
	Foot plantar pressure	Pressure field acting between the contact surface of the foot and the ground.	
EMG parameters	Electrical activities of lower limb muscles	Interpretation of the biological signals responsible for muscle activation.	[37]

2.3 General factors affecting gait

There are several factors that influence gait directly or indirectly. It is mostly dependent on factors such as gender, age, body weight and walking speed.

a. Gender

Men and women often have differences in their biomechanics, including muscle mass, joint flexibility and skeletal structure. These differences can result in variations in gait patterns [38]. Typically, men have longer strides and step lengths than women due to differences in leg length and muscle mass. This can affect the overall gait and walking speed. Women tend to have wider pelvises, which can influence the angle of their gait and affect the way they distribute their weight while walking.

b. Age

As people age, they may experience a decline in muscle mass and strength. Weaker muscles can lead to a slower and less steady gait. Age-related changes in joints can result in reduced joint flexibility, which can impact the range of motion during walking [39]. Balance and coordination tend to decline with age, making older individuals more susceptible to balance-related gait issues.

c. Walking speed

Faster walking speed is associated with longer strides and a more fluid gait. A slower pace often involves shorter steps and may result from factors such as fatigue or physical limitations. Variability in walking speed can be indicative of certain medical conditions or an increased risk of fall [40]. A steady and consistent walking speed is generally considered a sign of a healthy gait.

d. Body weight

Excess body weight places additional stress on the joints, particularly in the lower extremities. This can lead to altered gait patterns as individuals try to reduce joint discomfort. Higher body weight can affect balance and stability, potentially leading to a wider base of support during walking to maintain equilibrium [41]. Walking with excess body weight requires more energy and effort, which can result in a slower walking speed and a modified gait to conserve energy.

2.4 Techniques used for gait analysis

Several techniques have been mobilized in gait analysis so far. These techniques differ in terms of information they provide and methods they follow for gait analysis. Some of these are more useful in a research lab but less efficient in routine clinical practice. In context of Nepal, still traditional ways are used for gait analysis. Specialists observe the pattern of patient's gait simply by asking him/her to walk.

Goniometer has been used for the analysis of motion at a simple hinge joint such as elbow and knee joints for many years. It is commonly used by orthopedic surgeons and physiotherapists to measure joint angles [42]. Goniometers are in both short and long-arm forms but short-arm goniometer might not seem to provide accurate results when measuring joint angles around the knee due to the longer lever arm involved [43]. To improve accuracy, rotary electrical transducers can be attached to the goniometer's arm, transforming it into an electro-goniometer. This electronic component helps in precise angle measurements. But one of the challenges associated with this system is difficulty in positioning measurement devices properly over the human joint. So, it is no longer in use for kinematic gait analysis.

More advanced approach in kinematic gait analysis compared to the goniometer involves the optical tracking system, in which the computer is interfaced with a CCD camera. This setup tracks the movement of the markers positioned at specific anatomical sites on the limb segments [44]. However, this method is susceptible to inaccuracies that occur due to skin artifacts and incorrect placement of the markers. To address these concerns, markerless systems have been developed. This technique, driven by deep learning algorithms, utilizes synchronized video data to perform 3D human pose estimation [45]. Recent developments in gait analysis system have introduced libraries such as Openpose and PoseNet, which have shown potential in 2D multi-person pose estimation. These libraries utilize Convolution Neural Network (CNN) to identify the joint centers, even with a regular webcam [46]. It is important to note that the accuracy of such system is yet to be fully evaluated.

Kinetic measurement in gait analysis focuses on the force between the foot and the ground, which could be measured by instrumented section of floor known as force platform. When a person walks on the platform, the force sensors detect magnitude and direction of the force being applied. Since it is often found to be challenging to step onto the platforms naturally, typically the step length needs to be modified to align with predetermined orientation of force plates [47]. Mat type system was also introduced later but had similar

problem as that of the force platform. Researches are being conducted with the possibility of using pressure sensors embedded underneath the shoesole.

Many researchers have developed and used plantar pressure measurement system which is a great concern in clinical, sports and rehabilitation studies. Motha et al. demonstrated an instrumental rubber insole printed with an embedded pressure sensor for the detection and investigation of plantar pressure [48]. Recently, some commercial techniques stand out in the analysis of plantar pressure, body motion and for gait characterization. Baropodometric Platforms and Sensorized Insoles are commercially available for the evaluation of foot plantar pressure and patient's posture during walking and standing [49]. Force Sensing Resistor (FSR) is the cost effective solution for the determination of plantar pressure.

Although most research on the kinetics of gait has concentrated on Ground Reaction Force (GRF), certain amount of work has also been done in the limb acceleration. Inertial Measurement System (IMU), which is a combination of an accelerometer, gyroscope and magnetometer, is gaining popularity for the determination of linear acceleration and angular velocity of specific regions of the body. With IMUs, various possibilities for sensor placement exist and researchers have tried a number of approaches for gait-related studies, placing IMUs on the different body segments with the shank being the most widely preferred lower body segment for gait analysis closely followed by the foot [50]. Even the most precise calibration will yield a non-zero bias so the angle calculated by this system will be subject to drift [51]. IMUs are also being used for the kinematic gait analysis or the measurement of different joint angles.

Typically, gait analysis is done by combining more than one of the systems described earlier; it might make the use of markers or markerless videotaping system for kinematic gait analysis with any one of the kinetic gait analysis system. The modular system with a combination of an insole or a shoe-type pressure sensor and an IMU sensor attached to the ankle and foot [52] provided with many gait parameters has already been used. A system for analyzing the gait cycle of pedestrians using a modular system was developed by Nantawat Pinkam [53]. Although there are several techniques for gait analysis, research is still being conducted to identify new, cost effective and more precise systems.

2.4.1 Use of treadmill in gait analysis

Traditionally, gait analysis used to be performed over-ground usually so as to record ground reaction forces using force plates embedded on the floor along with body kinematics and surface electromyography (sEMG) simultaneously [54]. There are both pros and cons of

performing gait analysis during over-ground walking as well as on the treadmill. In case of overground walking, subjects normally do not face problems in pacing along the given path unlike in treadmill walking where they need to be familiar with the experimental setup [55]. Moreover, overground walking provides experimental subjects the further sophistication of choosing their natural walking speed [56], that is gait velocity with the most rational energy expenditure [57]. Even though such method of gait analysis has been efficient in generating clinically useful information, walking with one single step on a force plate, during the determination of ground reaction force, is less likely to resemble normal walking behavior in our daily life [58]. Also, multiple force platforms are required to determine force parameters in a number of consecutive steps which can be space consuming, more expensive and hence infeasible [59]. Similarly, this process requires a greater number of trials which eventually increases the data collection time. Therefore, treadmill-based gait analysis is being more popular nowadays.

A treadmill-based gait analysis is often used to overcome the limitations and constraints of overground walking. Treadmill walking offers several advantages since the pacing subject remains stationary on the treadmill and recording devices for electromyographic, kinetic and kinematic analysis can be applied effortlessly [60][61][62].One of the advantages of treadmill-based gait analysis is the elimination of the need for precise foot placement since every step is recorded. Thus, multiple cycles containing gait data can be recorded quickly, boosting the effectiveness of data gathering. As a result, more subjects can be experimented each day at a lower cost per individual for a clinical gait study [58]. Additionally, gathering data of multiple steps over the same space enhances its efficiency and reliability [63] thereby enabling the investigation of variability and changes over time. A treadmill also permits the use of a body weight support device or a safety harness. As a result, measurements can also be taken from patients with mobility problems. The ability to control and maintain walking speed is another benefit of using treadmill-based system [58].

Despite the fact that van Ingen Schenau (1980) claimed that the basic biomechanics of overground and treadmill walking are the same, potential differences in biomechanical, electromyographic and metabolic parameters are the matter of concern among researchers [64]. By reducing natural variability and complexity, treadmill walking produces a gait pattern that is more stable and predictable [65][66][67]. However, the potential demerits of using treadmill-based system can be the compromised quality of force recordings on a long and vibrating belt leading to lower accuracy, reliability and dynamic sensitivity as well as the need for a dedicated expensive treadmill [54].

2.5 Gait abnormalities and related disorders

Human gait depends on a complex interplay of major parts of the nervous, musculoskeletal and cardio respiratory systems [68]. Gait defines the general health condition of a person which may be influenced by age, sex, height, weight or other different socio-cultural factors [69].The prevalence of gait and balance disorders is prominent with increase in age, from around 10% between the ages of 60 and 69 years to more than 60% in those over 80 years [70]. Gait disorder arises due to neurological conditions (such as sensory or motor impairments), orthopedic problems (such as osteoarthritis and skeletal deformities) and medical conditions (such as heart failure, respiratory insufficiency, peripheral arterial occlusive disease and obesity). Some of the common forms of gait disorders are mentioned in the Table 2.2.

Table 2.2 Phenomenological classification of gait disorders [71]

Gait Disorders	Characteristics
Hemispastic gait	Unilateral extension and circumduction
Paraspastic gait	Bilateral extension and adduction with stiffness
Ataxic gait	Broad base and lack of coordination
Sensory ataxic gait	Cautious, worsening without visual input
Cautious gait	Broad based, cautious, slow, anxious
Freezing gait	Blockage, esp. on turning
Propulsive gait	Centre of gravity in front of body, festination
Astasia gait	Primary impairment of stance/balance
Dystonic gait	Abnormal posture of foot/leg
Choreatic gait	Irregular, dance-like, broad-based
Steppage gait	Weakness of foot extensors
Waddling gait	Broad-based, swaying, drop of swinging leg
Antalgic gait	Shortened stance phase on affected side
Vertiginous gait	Insecure, tendency to fall to one side
Psychogenic gait	Bizarre, rarely falls

The most common neurodegenerative diseases resulting in abnormal gait are Alzheimer's disease, multiple sclerosis, Parkinson's disease, Huntington's disease, cerebral palsy and dementia [70]. Osteoarthritis and skeletal deformities of the lower extremities are

the most common reasons for non-neurological gait disorders in adults characterized by a limited range of motion, avoidance of weight-bearing and asymmetry or limping [68]. Early detection of gait abnormalities in asymptomatic patients may infer to certain diseases that might occur in future. Hence, gait analysis is crucial in the detection and treatment plan of various diseases as well as clinical conditions.

Some of the common features of gait disorders observed in different diseases and clinical conditions are discussed as follows:

a. **Parkinson's disease:** Individuals with Parkinson's disease often experience a shuffling gait with small, rapid steps, reduced arm swing and difficulty initiating or stopping movements due to impaired motor control.

b. **Multiple Sclerosis (MS):** It can lead to various gait disturbances such as weakness, muscle spasms and problems with coordination and balance.

c. **Stroke:** A stroke can damage the brain's motor control areas, resulting in weakness, paralysis or impaired coordination that affects walking.

d. **Peripheral neuropathy:** Conditions like diabetic neuropathy can lead to sensory loss, muscle weakness and altered gait patterns due to damage to the nerves in the legs and feet.

e. **Muscle and joint disorders:** Conditions such as muscular dystrophy or arthritis can weaken muscles, affect joint mobility and lead to an abnormal gait.

f. **Spinal cord injury:** Damage to the spinal cord can result in paralysis or weakness in the lower extremities, leading to altered gait patterns.

g. **Orthopedic conditions:** Joint problems such as hip or knee osteoarthritis can cause pain and limitation of movement, which may affect a person's gait.

h. **Vestibular disorders:** Disorders of the inner ear's balance system, such as Benign Paroxysmal Positional Vertigo (BPPV), can lead to problems with balance and a staggering gait.

i. **Cerebral palsy:** This congenital disorder affects muscle control and coordination and often leads to a distinctive gait characterized by stiff, jerky movements.

j. **Medication side effects:** Some medications, especially those that affect the central nervous system, can cause dizziness, muscle weakness or changes in coordination, influencing a person's gait.

k. Aging: Natural aging processes can lead to muscle weakness, reduced joint flexibility and changes in balance, all of which can contribute to an altered gait.

l. Psychogenic gait disorders: In some cases, psychological factors can lead to functional gait disturbances that are not related to physical or neurological issues.

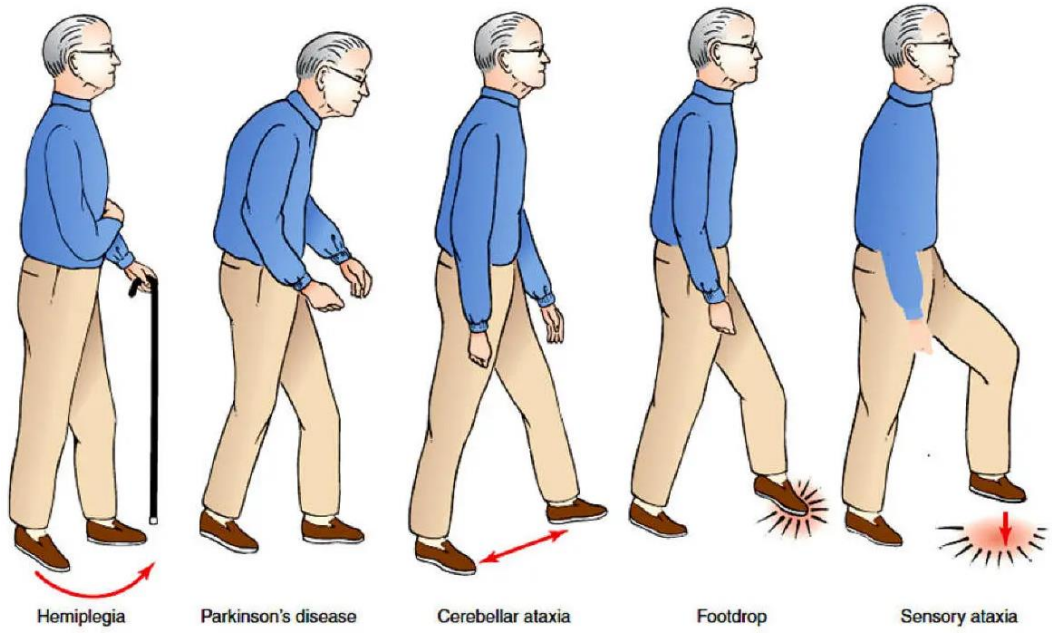


Figure 2.2 Patterns of various gait disorders [72]

2.5.1 Gait analysis for knee osteoarthritis

Osteoarthritis is a degenerative joint condition characterized by the breakdown of cartilage and the subsequent pain and stiffness in the affected joints that affects millions of people worldwide. Osteoarthritis mainly affects weight bearing joints such as hip, knee and spine with knee osteoarthritis being the most common one. It initiates impact on various joint tissues well before the middle age but its diagnosis occurs many decades later when the symptoms start to develop [73]. It is a progressive condition that could potentially result in disability, however the severity of clinical symptoms can differ from one individual to another [74]. Common symptoms of osteoarthritis include joint pain, stiffness, reduced range of motion, swelling and tenderness. Influence of the walking biomechanics on the progression of osteoarthritis can be better understood through gait analysis which also helps in the design of therapeutic interventions [75]. Some studies of gait analysis showed reduced gait speed, stride length and knee flexion angle in the patients of osteoarthritis as compared to the healthy subjects [76].

Chapter 3 Materials and Methodology

3.1 Materials

3.1.1 MPU sensor

MPU 6050 is a Micro Electro-mechanical System (MEMS) which consists of three-axis accelerometer and three-axis gyroscope. It helps us to determine velocity, orientation, acceleration, displacement and other motion-related parameters.



Figure 3.1 MPU 6050

3.1.2 FSR sensor

FSR UX 402 is used to measure the underfoot pressures during stride in order to kinetically evaluate human gait. It is essentially the resistor that alters its value based on how hard it is pressed. As load weight or pressure increases, FSR's electrical resistance falls.



Figure 3.2 FSR UX 402

3.1.3 Arduino Mega

Arduino Mega 2560 is a microcontroller board based on the ATmega2560. It has 54 digital input/output pins (of which 15 can be used as PWM outputs), 16 analog inputs, 4 UARTs (hardware serial ports), a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header and a reset button.

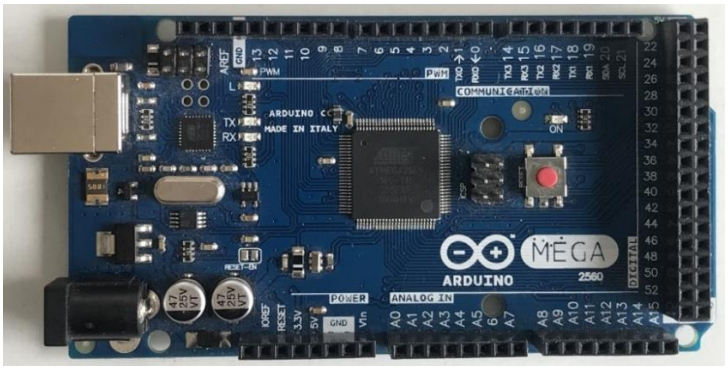


Figure 3.3 Arduino Mega 2560

3.1.4 Multiplexer

The TCA9548A I2C multiplexer allows us to communicate with up to 8 I2C devices using the same I2C bus. The multiplexer communicates with a microcontroller using the I2C communication protocol.



Figure 3.4 MUX TCA9548A

3.1.5 Bluetooth module

Bluetooth module HC-05 is a small range wireless communication device that can be used for various purposes like transmission of data or control of devices connected to it. Moreover, it can be interfaced with Arduino Mega and can be used where wireless communication is required for a small range.



Figure 3.5 Bluetooth module HC-05

3.1.6 Active markers

Active Markers (X-base: 14mm) are IR light emitting elements, mostly LEDs. All these markers emit radiation when the light source is sufficient to illuminate them from a considerable distance.



Figure 3.6 Marker (X-base: 14mm)

3.1.7 Velcro

Velcro is a brand name for a type of fastening tape consisting of opposing pieces of fabric, one piece with a dense arrangement of tiny nylon hooks and the other with a dense nylon pile, that interlock when pressed together. It can be used as a closure on garments, luggage, in place of buttons, zippers, etc.



Figure 3.7 Velcro

3.1.8 Ribbon cable

A ribbon cable is a cable with many conducting wires running parallel to each other on the same flat plane. It is used to provide connections between different hardware components on the developed system.



Figure 3.8 Ribbon cable

3.1.9 Shoe sole

A shoe sole is the bottom part of a shoe that comes into direct contact with the foot. The design and construction of shoe soles vary based on the individual size and intended use. It serves as a base for the placement of FSR sensors.



Figure 3.9 Shoe sole

3.1.10 Treadmill

Treadmill is a device consisting of a continuously moving belt, driven by a motor which is generally used for exercise. In gait analysis, it can be used to observe human gait patterns at different speeds with a number of step repetitions.



Figure 3.10 Treadmill WNQ model

3.2 Methodology

3.2.1 Overview of the system

The overall idea of our project can be illustrated as shown in the Figure 3.11 in relation to the assessment of gait parameters. Our system is basically comprised of the sensors (FSR UX 402 and MPU 6050) as well as the non-reflective active markers. After the assembly of these basic necessary components, the gait analysis system proceeds by performing two major operations simultaneously.

a. Video processing system
b. MPU and FSR sensors

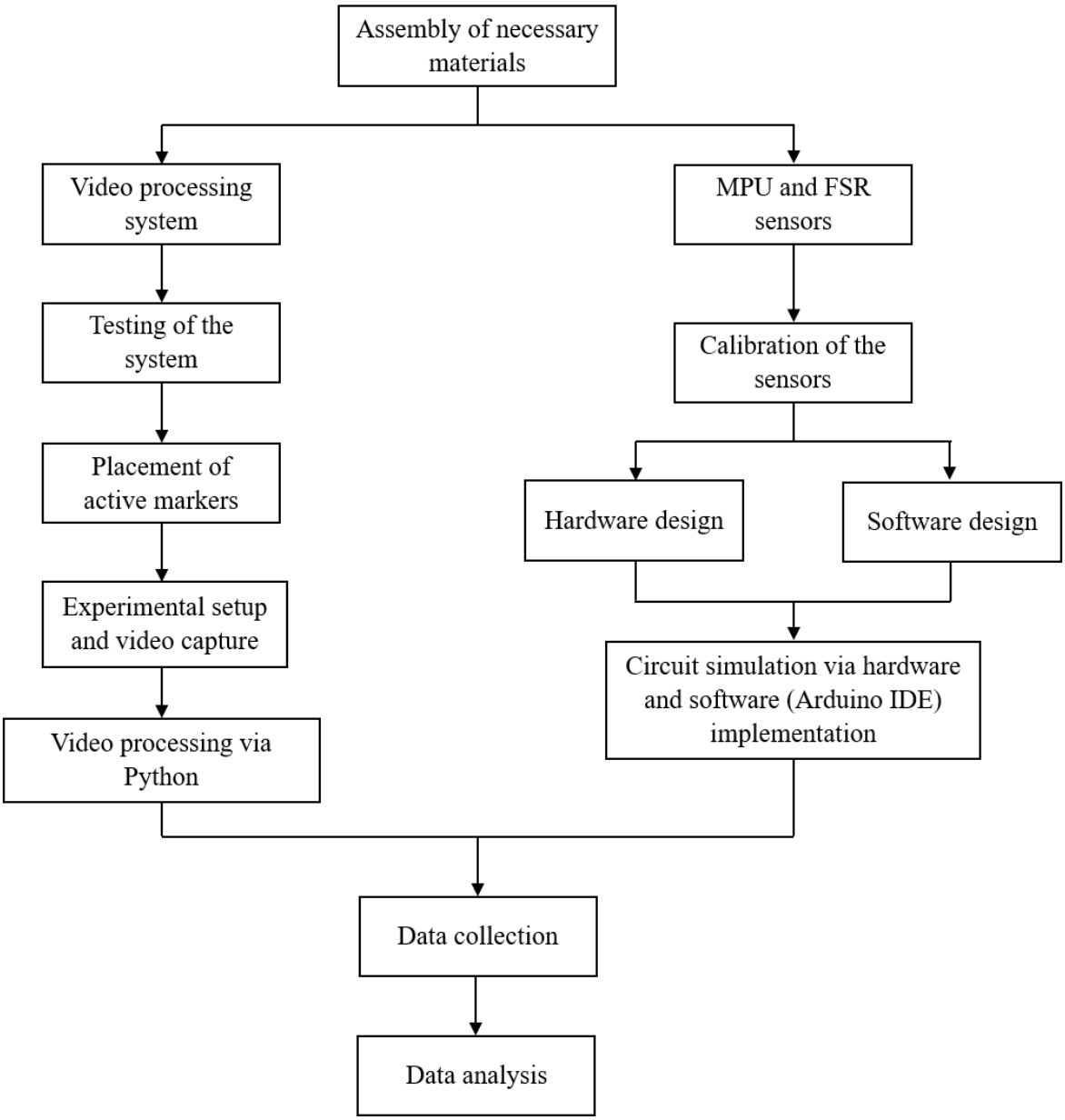


Figure 3.11 Block diagram showing overall methodology of the gait analysis system

a. Video processing system

The study of kinematic gait parameters such as rotation of motion of the hip joint, flexion and extension of the knee joint along with plantar flexion and dorsiflexion in the ankle joint in the sagittal plane is the main purpose of image processing in our project. For this, active markers were used in the localization of joints which could be captured in the camera system and then used for video processing followed by image processing for each frame to obtain the joint angles.

i. Testing of the system

Accuracy of the system was determined by manually visualizing and drawing hip, knee and ankle angles on the chart paper and then placing the markers at different locations, capturing the image of these markers followed by processing the captured image to determine these angles. The angles obtained from the image processing system were compared with those drawn manually on the chart paper as shown in the Table 3.1 so as to validate the efficiency of the system. The error percentage in each of these cases was found to be within 3%.

Table 3.1 Comparison of joint angles obtained manually and through image processing system

Joint angles	Joint angles measured manually	Joint angles calculated through image processing	Error percentage
Hip angle	53°	54.3°	2.45%
Knee angle	54°	54.56°	1.04%
Ankle angle	2.5°	2.57°	2.8%

ii. Placement of active markers

After testing the system, the active markers were placed in the anatomical landmarks such as shoulder, hip, knee and ankle joints as well as lateral forefoot position with the help of Velcro so as to create co-ordinate systems and subsequent joint centers. These landmarks were identified through expert physical palpation with the knowledge of anatomy and biomechanics.

iii. Experimental setup and video capture

The camera was set up at a suitable distance from the treadmill so that each marker was visible on the video throughout the motion capture of the subject. A black curtain was used in the background to ensure efficient detection of the landmarks via illumination of the markers through the flash light from the camera. After this setup, the video was captured during different gait phases of the experimental subject pacing on the treadmill.

iv. Video processing via Python

The captured video was then processed using Python Open CV library. During video processing, the code was implemented so as to calculate and represent the joint angles of different locations using the video file of pacing experimental subject on the treadmill.

During video processing, each of the frames followed the basic steps of the image processing as shown in the Figure 3.12.

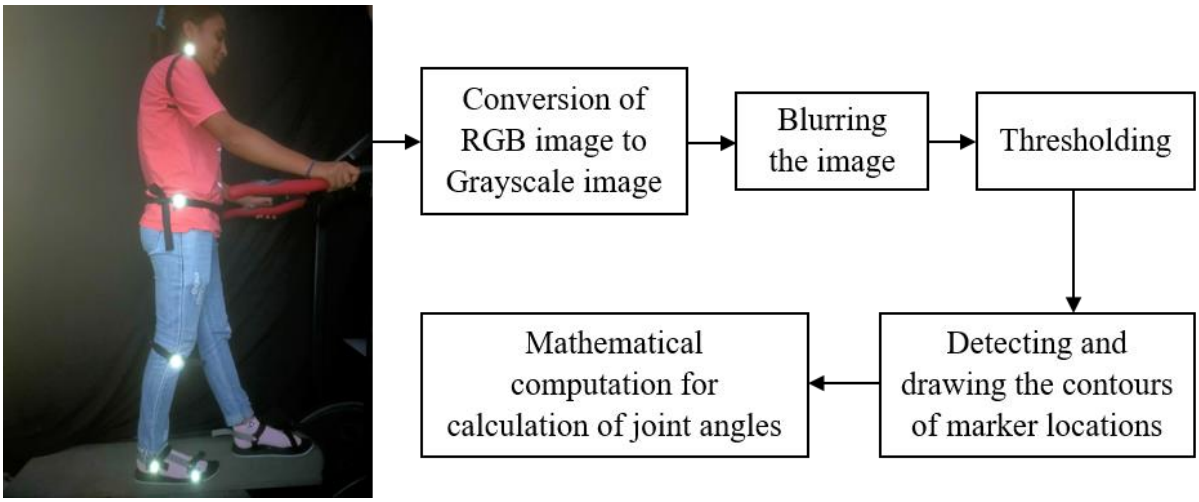


Figure 3.12 Image processing steps involved in each video frame during video processing

Initially, the image in each frame was read for image processing in the Python. This image was then changed into grayscale image which was further blurred followed by thresholding. After determining the contours of the illuminating markers, the midpoints of these contours were calculated and located which provided us with the specific coordinates to work with. These coordinates were used to determine the slope of the lines joining them using two-point slope formula, thereby calculating the angles using the formula given by equation (3.1).

$$\theta = tan^{-1}\left(\frac{m_1 - m_2}{1 + m_1 * m_2}\right) \dots\dots\dots\dots\dots\dots (3.1)$$

For the calculation of the hip, knee and ankle angles, the image processing system made the use of segment angles, i.e. trunk (θ_{tr}), thigh (θ_{th}), leg (θ_{lg}) and foot (θ_{ft}) angles with reference to the Canadian Society of Biomechanics (CSB) standard [77]. These segment angles were measured considering the horizontal reference and using the equation (1). The segment angle is the angle made between the body segment and the right horizontal of the distal end. It should be consistently measured in the same direction from a single reference; either horizontal or vertical. In this study, since the horizontal reference was considered, so the trunk is supposed to be 90° for normal human gait. Finally, the hip (θ_h), knee (θ_k) and ankle (θ_a) angles were calculated according to the expressions given in the Figure 3.13.

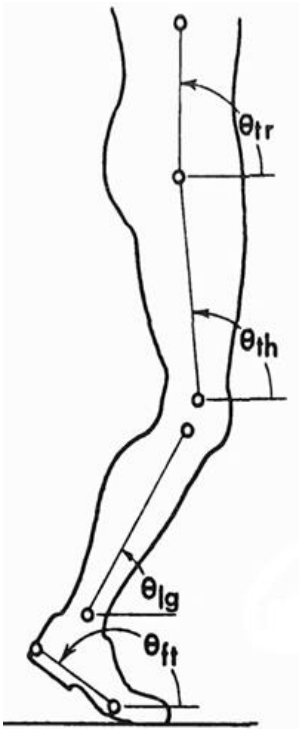


Figure 3.13 Representation of segment angles as per the CSB gait standard [78]

As per the CSB standard, the joint angles were calculated using the equations (3.2), (3.3) and (3.4) involving the segment angles.

$$\theta_h = \theta_{th} - \theta_{tr} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (3.2)$$

$$\theta_k = \theta_{th} - \theta_{lg} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (3.3)$$

$$\theta_a = \theta_{ft} - \theta_{lg} - 90° \ldots\ldots\ldots\ldots\ldots\ldots (3.4)$$

Here,

+ve angle for flexion / dorsiflexion

-ve angle for extension / plantar flexion

The general video processing system was executed as displayed on the flowchart in the Figure 3.14.

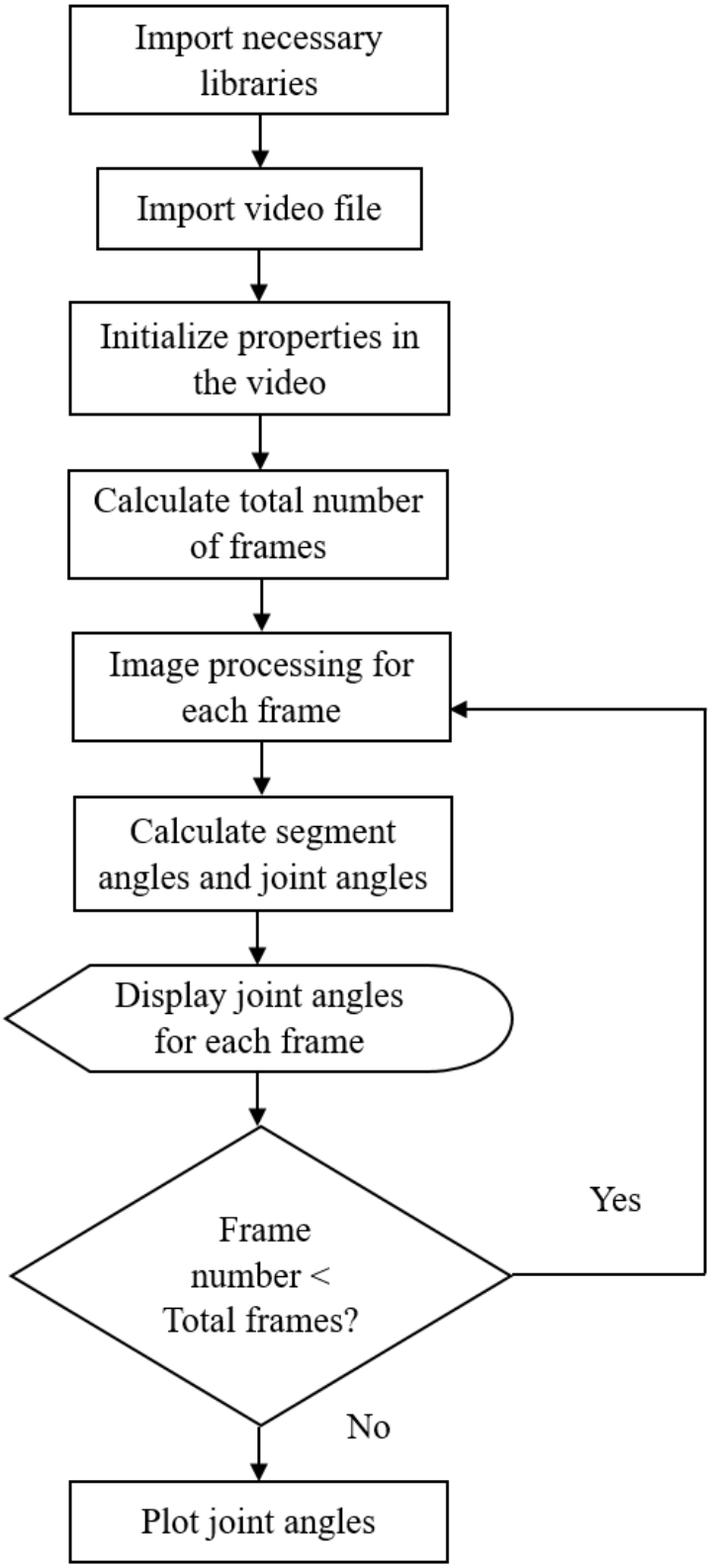


Figure 3.14 Flowchart showing the steps involved in video processing

b. MPU and FSR sensors

MPU sensor is basically used to track the movement and orientation of the body segments during gait. MPU, which consists of accelerometer and gyroscope, was placed on either side of thigh, shank and foot regions to provide information on acceleration and angular velocity of those regions. Likewise, FSR sensor was attached to the sole of the shoe to measure the force exerted by the foot on the ground. Basically, FSRs were placed on the heel, metatarsals and toe of each foot.

i. Calibration of the sensors

Before incorporating sensors for the development of the system, calibration of each of these sensors was done individually. MPU sensor, that provides acceleration and angular velocities along X, Y and Z direction, was calibrated by keeping the sensor still in any horizontal or vertical position and performing the proper test code for calibration. The raw data for acceleration (±2g) in any direction was found to be in proximity to one when the MPU sensor was directed in that particular direction whereas the data showed near to zero values for the remaining two directions, indicating the successful calibration of the MPU sensor.

On the other hand, FSR sensor was calibrated by applying varying range of forces to the sensor and noting their corresponding resistance values. These resistance values were further converted into respective conductance values following their inverse relationship.

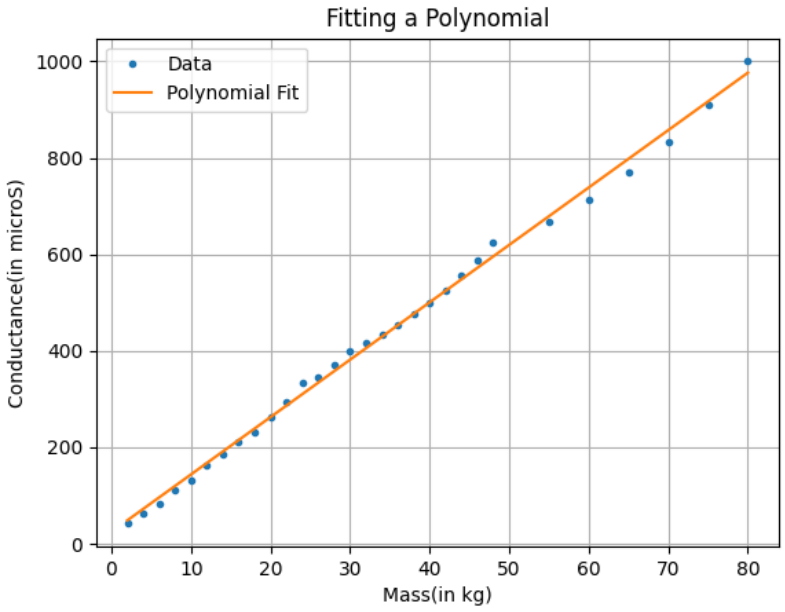


Figure 3.15 Plot of mass vs. conductance with a fitted straight line

Then, the graph was plotted between the various range of masses (in kg) and their respective conductance values (in microS) as depicted in the Figure 3.15 so as to determine the nature of relationship between the mass and conductance which can be represented by equation (3.5).

$$Calibration\ mass\ (kg) = \frac{Conductance - 24.979}{11.892} \ldots\ldots\ldots\ldots\ldots\ldots (3.5)$$

ii. Hardware design

After the successful calibration of the sensors, the MPU sensors were connected to the Arduino Mega via ribbon cable attached to the matrix board. Multiplexer TCA9548A was used to interface these six MPU sensors to the single Arduino Mega. Likewise, FSR sensors attached to the sole of the shoe were connected directly to the Arduino Mega via ribbon cable. Then data from the Arduino was transmitted to the laptop via Bluetooth module. The layout of the hardware connection can be represented in the block diagram as shown in the Figure 3.16.

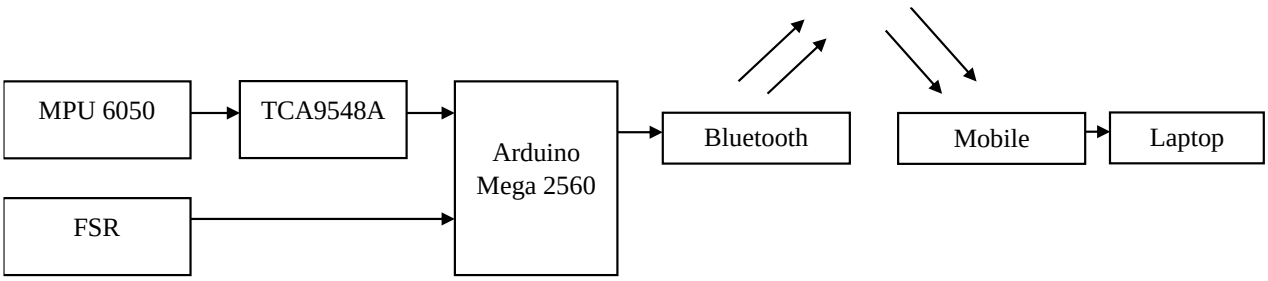


Figure 3.16 Block diagram showing MPU 6050 and FSR interfacing to Arduino Mega 2560

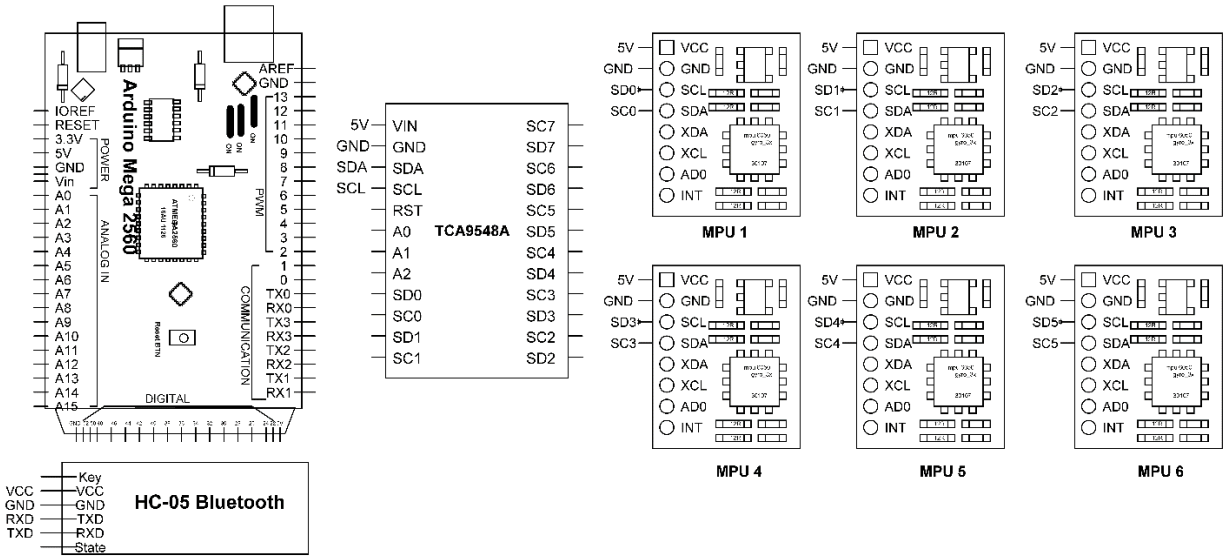
iii. Software design

Using the software tool Arduino IDE, the FSR sensors, MPU sensors along with the multiplexer, Bluetooth module and Arduino Mega were programmed to measure force exerted by the different regions of the foot as well as motion-related parameters of the joints. FSR sensor was interfaced with Arduino Mega and coded in such a way that the analog reading gets converted into digital form to get the resistance reading that eventually displays the force applied by the foot on the ground at different instances. Also, the MPU sensor interfacing was done so as to display the acceleration readings along different directions of the motion of the sensor.

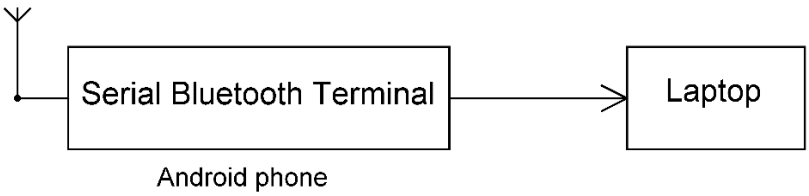
iv. Circuit Simulation

The hardware components and software were integrated properly so as to obtain the desired output in relation to assessment of the gait parameters and for further data collection and analysis. Data obtained via Bluetooth module in android mobile with the use of Serial Bluetooth Terminal were transferred to laptop for the data analysis purpose.

Figure 3.17 shows the circuit layout for the connection of six MPU sensors to the Arduino Mega via multiplexer in the transmitter side as well as acquisition of data in the receiver side.



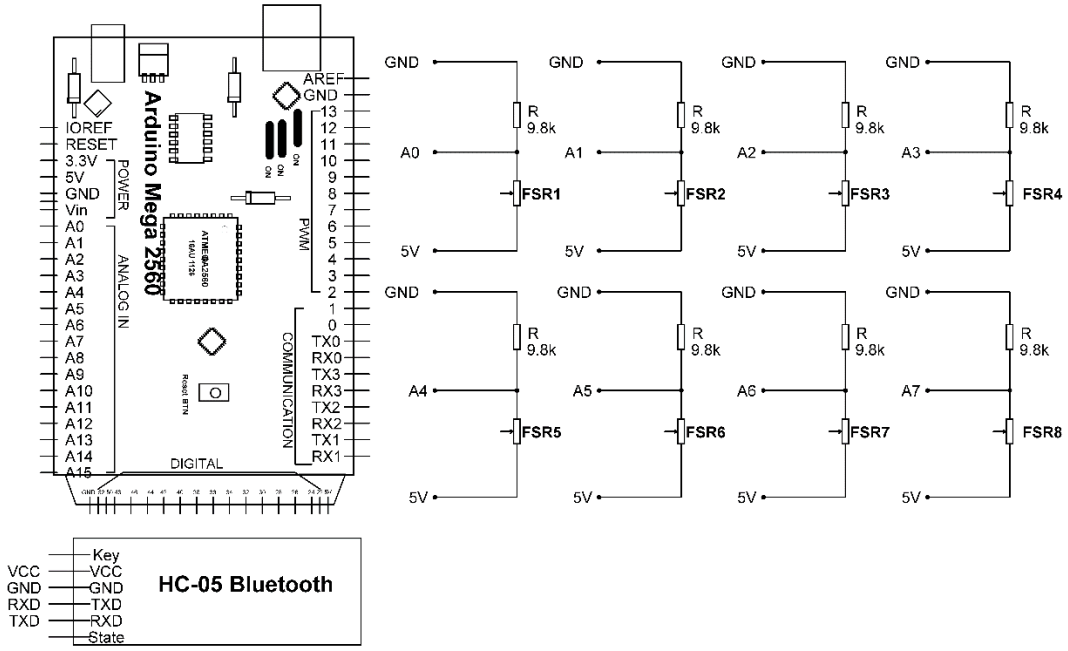
(a)



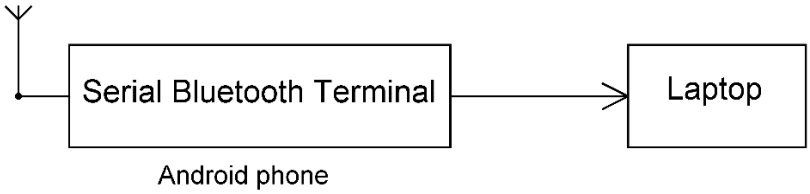
(b)

Figure 3.17 Circuit diagram involving MPU sensors for (a) transmitter side and (b) receiver side

Similarly, Figure 3.18 shows the circuit layout for the connection of eight FSR sensors to the Arduino Mega in the transmitter side as well as acquisition of data in the receiver side.



(a)



(b)

Figure 3.18 Circuit diagram involving FSR sensors for (a) transmitter side and (b) receiver side

Algorithm

a. For MPU 6050 sensor reading

- Initialize MPU 6050 sensor object.
- Set up full-scale range for the accelerometer reading.
- Set up a unique I2C address for the sensor.
- Calibrate the sensor.
- Enable the channel for each sensor.
- Read acceleration values from each sensor.
- Convert raw values to actual units using the appropriate scaling factors.
- Output the values to the Serial Monitor.
- Output the values to Bluetooth using the SoftwareSerial library.

b. For FSR sensor reading

- Define an array for the FSR pins.
- Define the resistor value, gain and offset for the FSR circuit determined through the calibration process.
- Read analog value from each FSR pin.
- Calculate resistance and conductance for each FSR pin.
- If the resistance value is greater than 23600, set the force to 0.
- Otherwise, perform calculation for the mass and force of the FSR sensor.
- Output the force values to the Serial Monitor and to the Bluetooth Module.

c. For Bluetooth module

- Initialize the SoftwareSerial library with the appropriate RX and TX pins.
- Output the sensor values to Bluetooth using the BTSerial object.
- Flush the BTSerial buffer to ensure all data are sent.

After the completion of these two sections, our project finally advanced for data collection and data analysis.

3.2.2 Data collection

Following the successful testing and implementation of the system, the data collection of the subjects was done in the experimental setup. For this, the experiment was conducted among 15 subjects with the normal gait and 3 patients with the knee osteoarthritis condition. Each of these subjects were asked to pace on the treadmill at speed 0.8 km/hr three times in order to collect data from the markers-based video processing system, MPU sensors and FSR sensors that determine the joint angles, motion-related parameters and force readings respectively. These data obtained from the video processing of the subjects as well as from Arduino Mega via Bluetooth module were extracted in the Excel sheet so as to utilize them for the analysis.

3.2.3 Data analysis

Data analysis is done in order to derive some relevant information about the gait-related parameters of the subjects from the developed system. In our project, the data related to joint angles obtained from video processing system, force exertion from FSR sensors and acceleration values from MPU sensors were collected and analyzed in the MS-Excel. The

hip, knee and ankle angles for both normal and abnormal subjects were precisely identified and located for each gait phase and these joint angles were plotted for one complete gait cycle. Similarly, the force exertion was obtained, analyzed and the plotted between each region of the right and left feet for both normal and abnormal subjects to study the variation of force distribution in each case. Finally, the nature of motion was studied from the data of MPU sensors in each of the thigh, shank and foot regions for these subjects. In general, this analysis provides the framework for comparison of gait-related parameters between the normal and abnormal subjects, particularly the ones with the conditions of knee osteoarthritis.

Chapter 4 Results

4.1 Video processing system

The successful implementation and testing of the video processing system enabled us to perform computation and analysis of the joint angles during each gait phase for all of the 15 normal subjects and 3 patients with knee osteoarthritis.

The comparison of joint angles has been drawn between one of the 15 normal subject group and one of the 3 patient group with condition of knee osteoarthritis, both of which weigh about 55-60 kg and measure around 155-165 cm in height. Each of the knee, hip and ankle angles were computed for both of these groups during individual phases of a complete gait cycle which were obtained as listed in the Table 4.1.

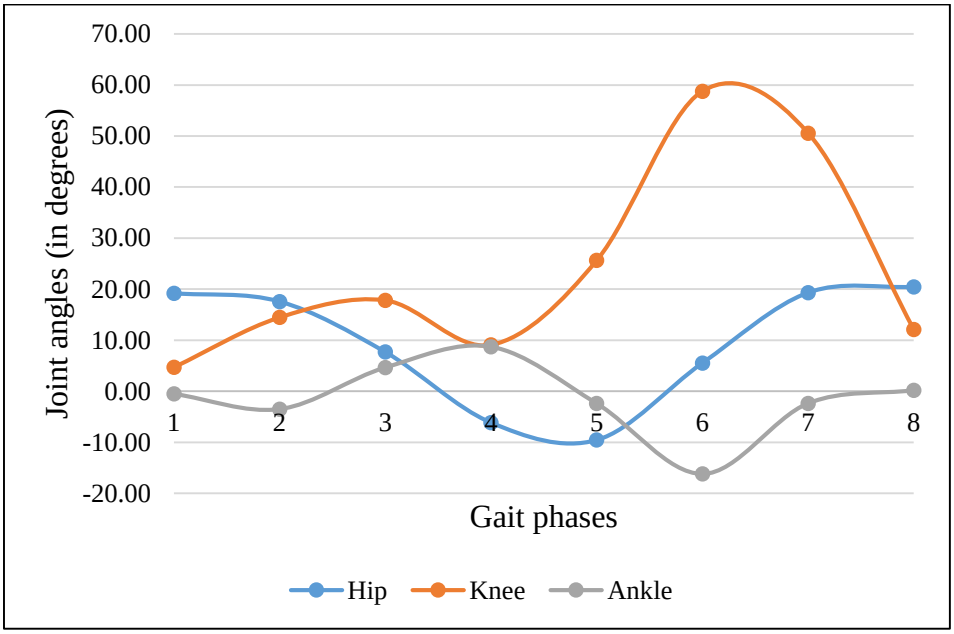
Table 4.1 Comparison of joint angles between the normal and abnormal subject groups during different gait phases

Gait Phases	Normal Subject			Abnormal Subject		
	Hip angle	Knee angle	Ankle angle	Hip angle	Knee angle	Ankle angle
Initial Contact (2%)	13.63	10.20	-8.46	3.77	-3.46	-2.75
Loading Response (12%)	11.45	11.36	-6.27	1.14	-2.21	-1.93
Mid Stance (31%)	7.39	13.06	0.56	-0.24	-0.86	1.81
Terminal Stance (50%)	2.64	13.71	7.72	-3.96	4.65	8.86
Pre Swing (62%)	-1.68	23.25	12.00	-2.70	25.86	3.28
Initial Swing (75%)	10.95	54.56	-1.20	11.00	17.53	4.91
Mid Swing (87%)	22.33	46.91	3.69	15.88	-0.16	2.34
Terminal Swing (100%)	19.42	8.84	-4.76	9.38	-2.70	0.52

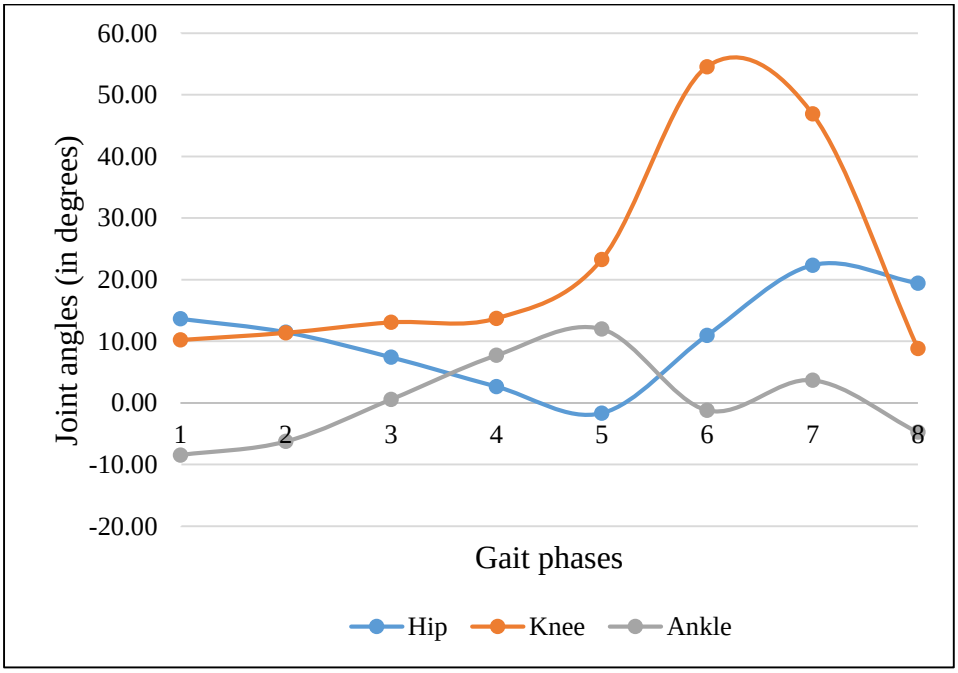
*All the angles are in degrees.

From the table, we can observe the significant differences in all of the joint angles during various gait phases among these two groups. The angles have transitioned sharply and to a significant level from one phase to another phase of the normal subject group whereas the

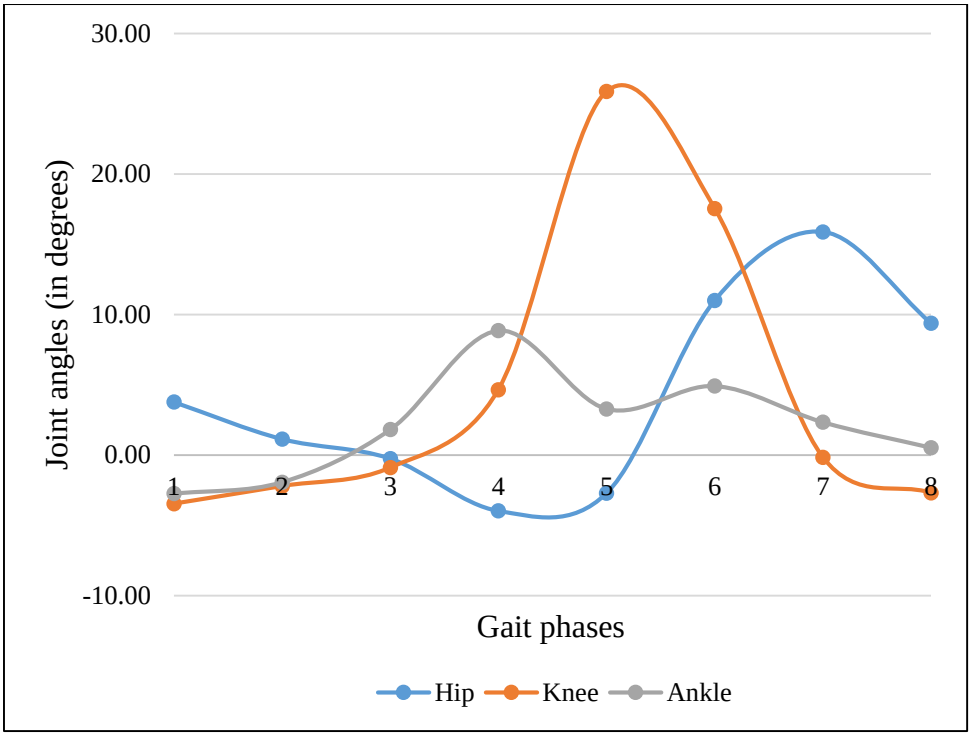
transition is much gradual in case of the abnormal subject group. The standard graphical representation of the hip, knee and ankle angles for the normal individual during one gait cycle is depicted in the Figure 4.1 (a). From the graph, it is clear that the standard for the hip angle ranges from -10° to 20°, the knee angle from 0° to 60° and the ankle angle varies from -20° to 10°. Similarly, these joint angles for one complete gait cycle can also be represented graphically in cases of both of these subject groups as shown in the Figure 4.1 (b) and (c).



(a)



(b)



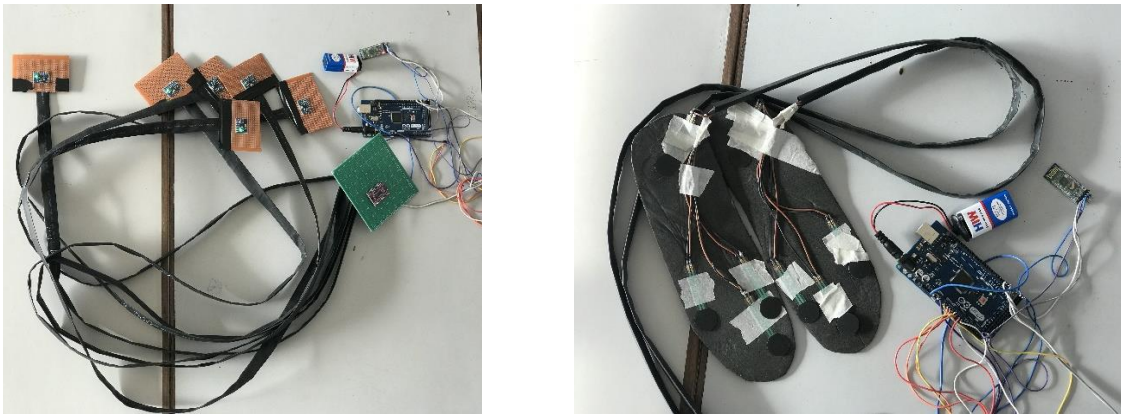
(c)

Figure 4.1 Graphical representation of joint angles during one complete gait cycle for (a) standard reference (b) normal subject group and (c) abnormal subject group

In all of these cases of the Figure 4.1, the numbers 1-8 represent the gait phases: initial contact, loading response, mid stance, terminal stance, pre swing, initial swing, mid swing and terminal swing respectively. The graph of the normal subject group exhibits the variation of the hip angle from -1° to 22°, the knee angle from 10° to 55° and the ankle angle from -9° to 12°. Similarly, in case of the abnormal subject group (i.e. patient group with knee osteoarthritis), the hip angle ranges from -4° to 16°, the knee angle from -3° to 26° and the ankle angle from -3° to 9°.

4.2 Wearable sensors

Following the calibration of the FSR and MPU sensors as well as the implementation of their proper circuit connections including interfacing of the multiple sensors to the Arduino Mega, the overall system for gait analysis was successfully developed. The system was designed and developed as shown in the Figure 4.2. The Figure 4.2 (a) shows the connection of six MPU sensors through multiplexer to the Arduino Mega whereas the Figure 4.2 (b) shows the connection of eight FSR sensors to the Arduino Mega, both of the MPU and FSR sensors which are finally connected to the Bluetooth module.



(a) *(b)*

Figure 4.2 Development of (a) MPU interfaced system and (b) FSR interfaced system

All the experimental subjects from both normal and abnormal groups were made to walk on the treadmill twice at a predetermined pace with the developed system fixed on the proper locations of their body, once with the FSR interfaced system and next time with the MPU interfaced system. Finally, the respective force exertion data and acceleration data from the FSR and MPU sensors were obtained and recorded in the separate Excel sheets.

For the normal group of 15 subjects with height 159.17±5.30 cm and weight 50.13±5.25 kg, one complete gait cycle was completed in 1.56±0.20 seconds whereas for the abnormal group of 3 patients with height 159.17±7.76 cm and weight 58.67±4.04 kg, the time required for completion of one gait cycle was found to be 1.84±0.45 seconds.

4.3 Output of the sensors

4.3.1 FSR sensors

Data from the FSR sensors indicate force applied at toe, first metatarsal, fifth metatarsal and heel regions of each foot of the experimental subject groups. These force exertion values at each region of both right and left feet were simultaneously plotted in order to compare the nature of force application at those respective regions between the two feet during various gait phases while pacing on the treadmill.

The graphical representation of the force exertion at toe, first metatarsal, fifth metatarsal and heel regions of the right and left feet for both the normal and abnormal subject groups was observed as shown in the Figure 4.3, Figure 4.4, Figure 4.5 and Figure 4.6 respectively.

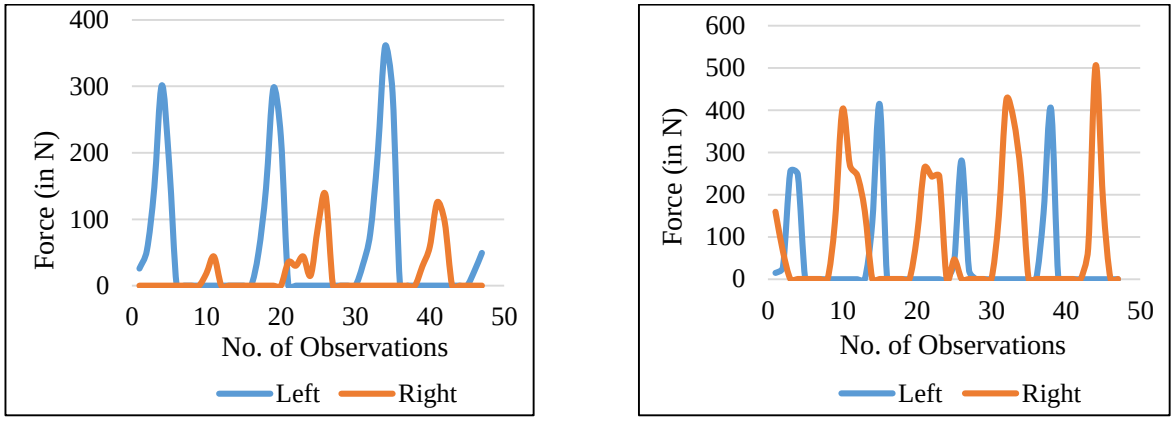


Figure 4.3 Force distribution of right and left toes for (a) normal and (b) abnormal subject groups

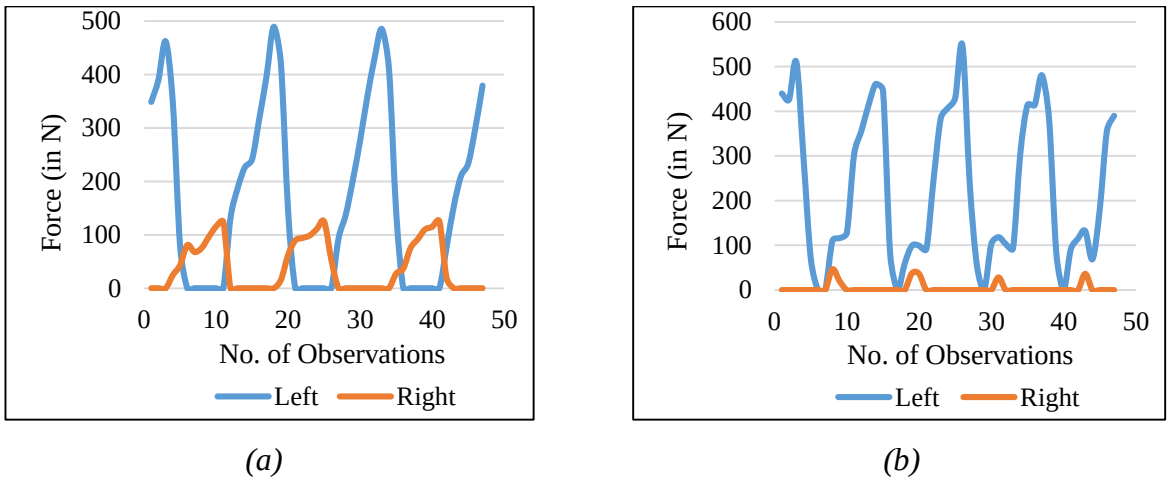
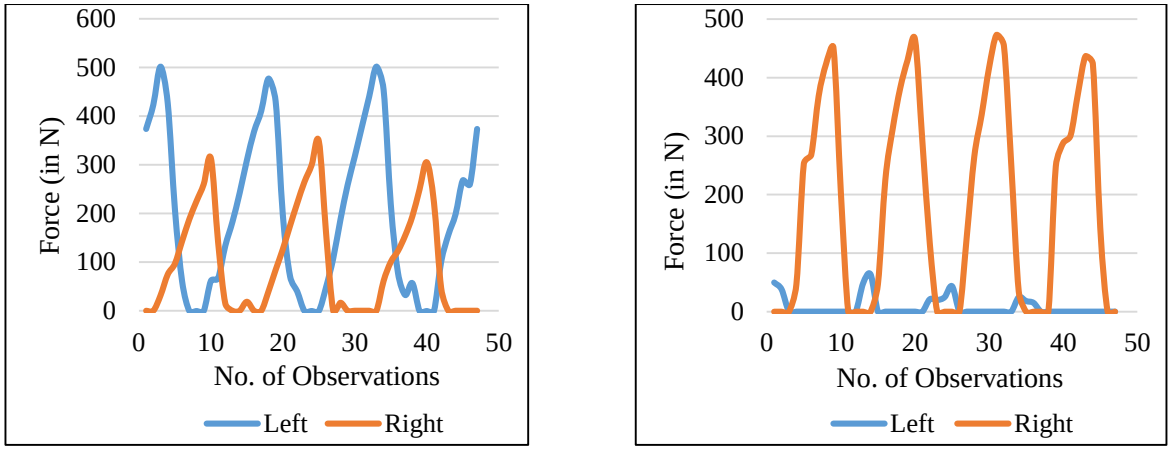
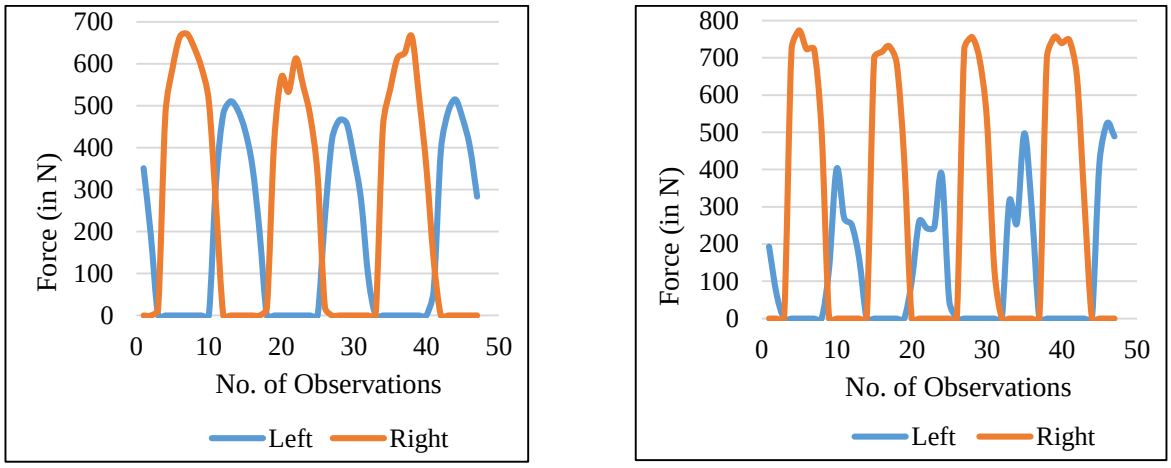


Figure 4.4 Force distribution of right and left first metatarsals for (a) normal and (b) abnormal subject groups



(a) *(b)*

Figure 4.5 Force distribution of right and left fifth metatarsals for (a) normal and (b) abnormal subject groups



(a) *(b)*

Figure 4.6 Force distribution of right and left heel regions for (a) normal and (b) abnormal subject groups

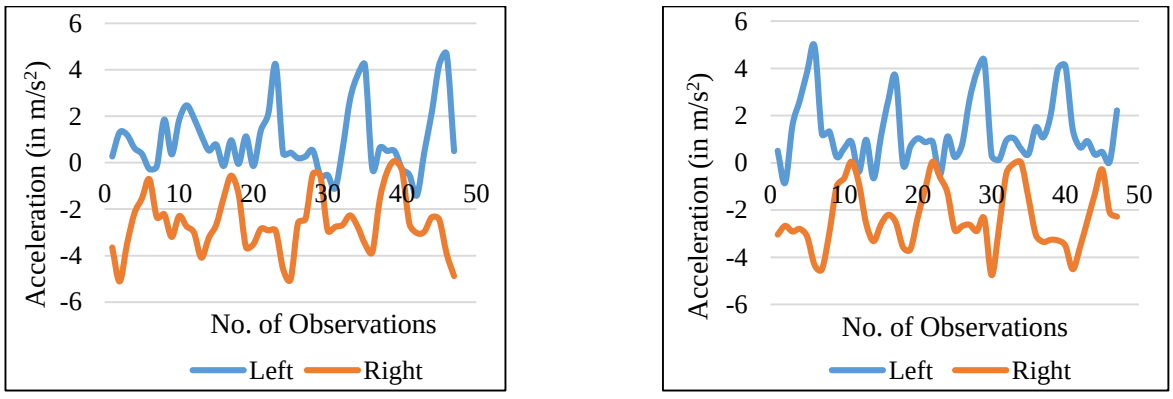
In each of these figures, 50 data were taken for each subject group to plot the graphs of force exertion at each region of the foot for a number of gait cycles. These graphs represent the patterns of force distribution at different regions for the normal and abnormal subject groups. As depicted in the figures, all these regions of the right foot of the abnormal group cover more number of cycles as compared to the normal group. In normal subject group, the amplitude of force exertion almost appears as maximum in the heel, average in the metatarsals and minimum in the toe. The force distribution among the right and left feet is not perfectly balanced owing to the fact that the individual can apply unequal pressures between the two feet during normal walking. While the force application in the right and

left heel regions of the normal group is comparable, the force exertion in the right heel of the abnormal group is dramatically significant compared to that in the left heel suggesting that the problem lies in the left leg of the patient.

4.3.2 MPU sensors

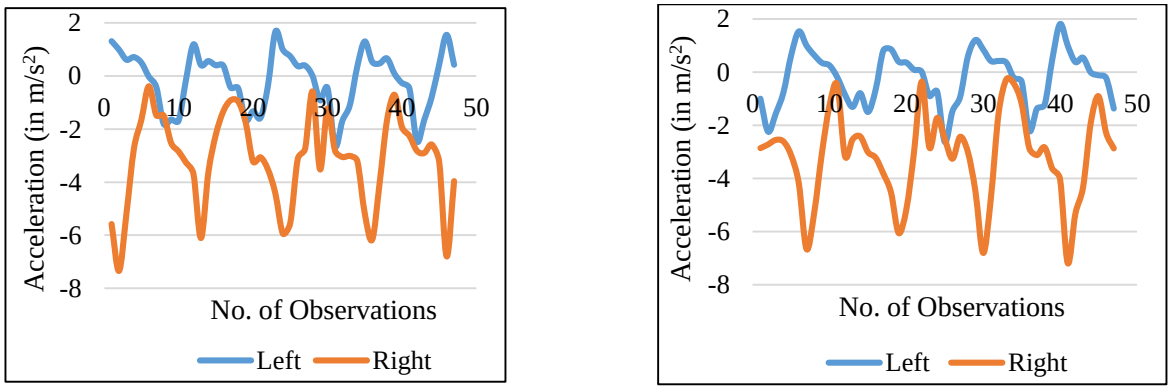
Again, these experimental subjects from both normal and abnormal groups were made to walk on the treadmill with the developed MPU interfaced system fixed on their lower extremities to record data from the MPU sensors at thigh, shank and foot regions of each leg. These acceleration values from each region of the two legs were plotted simultaneously.

The nature of motion at thigh, shank and foot regions of right and left legs for the normal and abnormal subject groups was obtained as shown in the Figure 4.7, Figure 4.8 and Figure 4.9 respectively.



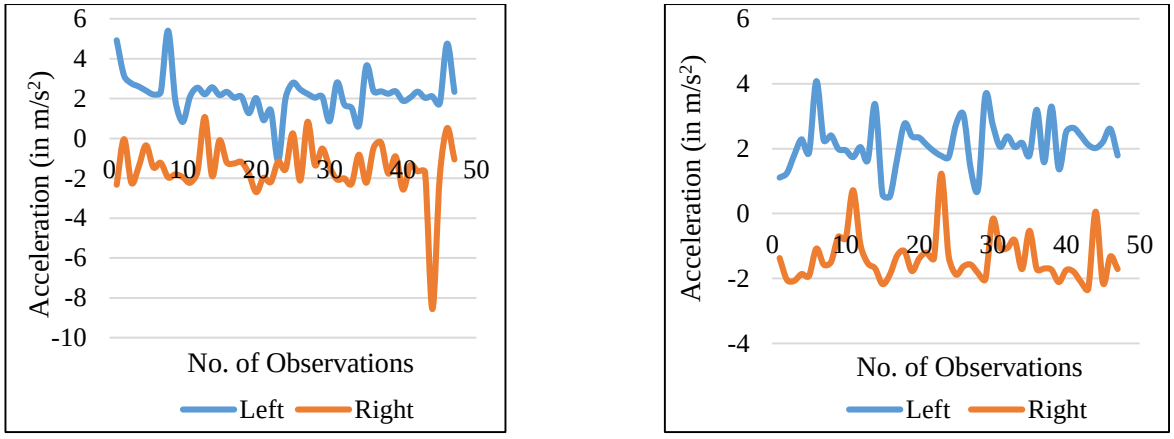
(a) *(b)*

Figure 4.7 Nature of motion of right and left thigh regions for (a) normal and (b) abnormal subject groups



(a) *(b)*

Figure 4.8 Nature of motion of right and left shank regions for (a) normal and (b) abnormal subject groups



(a) *(b)*

Figure 4.9 Nature of motion of right and left foot regions for (a) normal and (b) abnormal subject groups

Similar to the graphs obtained in the force distribution, in each of these figures as well, 50 data were taken for each subject group to plot the graphs at each region of the lower extremities for a number of gait cycles. These graphs represent the nature of movement at different regions for the normal and abnormal subject groups. As depicted in the figures, all these regions of both the normal and abnormal groups exhibit regular patterns of the right and left legs. While the nature of motion of thigh and shank regions of the abnormal group is almost similar to that of the normal group, it appears to be slightly varying in the foot region as compared to that of the normal group.

Chapter 5 Discussion

Our project is basically designed to develop a system for gait analysis comprising of active markers to allow measurement of joint angles through camera kinematic system along with the MPU sensors for obtaining acceleration of the lower extremities and the FSR sensors for determining the force exertion on different regions of the foot.

As a result of markers based video processing system, a specific range of hip, knee and ankle angles were obtained for all of the experimental subjects from both groups. The normal individual group exhibit the variation of joint angles in the range comparable to the previous studies of the healthy subjects walking on the treadmill [79]. However, the joint angles of some of the normal subject group fall outside this range owing to the fact that the slight misplacement of the active markers can significantly alter the value of joint angles. Inconsistent marker placement has been shown to contribute up to 3% error in the lower joint angles during testing procedure of the image processing system.

When comparing the joint angles between the normal and abnormal subject groups, significant differences could be observed. Starting with the hip angle, the normal group exhibited a broader range of motion (-1° to 22°) compared to the abnormal group (-4° to 16°). This suggests that the abnormal group may experience restrictions in their hip movement, potentially affecting their overall gait pattern. In case of the knee angle, the normal subject group again demonstrated a larger range of motion (8° to 55°) in comparison to the abnormal patient group (-3° to 26°). This indicates that the abnormal group may have reduced flexion or extension capabilities of their knees during the gait cycle. The designed gait analysis system is accurate enough for clinical assessment of knee movements to identify knee diseases. Similarly, for the ankle angle, the abnormal group further displayed a narrower range of motion (-3° to 9°) when compared to the normal group (-8° to 12°). This implies that individuals with abnormal gait patterns may observe limitations in their ankle joint movement during normal walking. Altered ankle motion can impact stability, propulsion and weight distribution throughout the gait cycle. This marker based joint angle detection system can only provide approximate guidance in certain situations, such as surgical procedures and biomechanical experiments involving intact motion, due to limitations caused by soft tissue artifacts [80]. To mitigate the effects of soft tissue artifacts in gait analysis, invasive markers in the form of bone pins can be employed. By directly

attaching these markers to the bones, the influence of soft tissues on the accuracy of the measurements can be minimized [81][82].

The sensors based system was developed by sticking to the algorithm specified in the methodology section and ensuring correct connections between the different hardware components along with their proper software implementation in addition to the successful calibration of the sensors. This system is capable of providing output for the study of kinematics as well as kinetics parameters with the use of MPU as well as FSR sensors respectively. We tried to design and generate this system as manageable and as systematic as possible to make the data collection process easier and more efficient. However, during the development of the system, all the MPU sensors were provided the same address for their connection to the Arduino Mega which made it difficult to distinguish between the sensors placed at different regions of the lower extremities. So, we made the use of the multiplexer to provide six separate addresses to these six sensors and differentiate the data acquired from each sensor.

Based on the data of force distribution obtained from the FSR sensors in the four different foot regions, comparison could be drawn between the normal and abnormal subjects. Over a 5 second walking phase, the nature of force distribution on the right and left feet of the normal group almost follows as: maximum at heel region followed by first and fifth metatarsals with minimum force at toe region. This force distribution is comparable to the previous studies showing amplitude graph of the foot pressure patterns of the normal gait [83]. Highest peak forces are found underneath the heel areas, due to the large impact peak forces occurring during initial contact phases of barefoot running [84]. The heel, being the first point of contact with the ground, experiences the brunt of impact forces, leading to the highest recorded peak forces in the heel region in both normal and abnormal gait cases. However, this amplitude variation at different pressure points was not observed the same for all the cases of the normal group. It is important to acknowledge that these distributions vary among different experimental subjects, primarily due to difference in the body mass and the placement of FSR sensors in this specific setup [85]. For the abnormal patient group with knee osteoarthritis, maximum force exertion was observed at pressure points of the right foot as the problem lied in their left leg.

In case of MPU sensor integrated system, lateral aspect of foot over the base of the metatarsal, shank and thigh regions were chosen as this location is a flexible part of the foot during motion [86]. From the data of acceleration values obtained by interfacing six MPU sensors with the multiplexer, we observed the nature of motion at different regions of the

lower extremities of both the normal and abnormal subjects. Almost all of these subjects demonstrated similar nature of movement alterations in their treadmill waking. The normal and abnormal subjects demonstrated similar patterns of movement at these regions except for the foot region. This study also quantifies the progressive transfer of accelerations between multiple adjacent body segments (i.e. foot-shank-thigh) [87]. This transfer of accelerations provides valuable insights into the transmission and coordination of movements within the lower extremities. It provides deeper understanding of biomechanical interactions and load distribution within the kinetic chain during normal walking.

Although this combined system with camera kinematic system along with the MPU and FSR sensors offers a thorough approach to gait analysis, it contains some drawbacks that need to be addressed. One of these drawbacks is the power supply limitations. Depending on the power source, the system has limitations in terms of battery life or power capacity, potentially affecting the duration of data collection sessions. Another is the challenge in managing a heavy wire system during data collection. The presence of multiple wires connecting the various components became cumbersome and interfered the natural movement of the subjects, potentially impacting the accuracy of the collected data. Additionally, the soldering that sustains the weight of a prototype gait analysis system using ribbon cable wires could be of some concern. If the soldering is not secure and strong enough or if the weight of the system is not adequately supported, it could lead to instability or even failure during data collection, resulting in incomplete or unreliable data.

Likewise, other limitation includes the requirement for a dark room for marker detection in the designed gait analysis system. This is because the markers used in the system need to be illuminated and accurately detected by the camera. The need for a controlled lighting environment can be inconvenient and limit the flexibility of data collection, as it requires a specific setup and may not be feasible in certain settings. However, despite these limitations, by addressing the power supply issues using power bank, improving the wire management, ensuring robust soldering and setting up the dark room, the system was properly utilized. Through fixing these minor issues and conducting thorough evaluations of the system's efficiency, the data collection and analysis of the experimental subjects have been conducted successfully.

Chapter 6 Conclusion and Further Works

In a nutshell, the overall system of gait analysis involves a combination of key components such as video processing system, MPU and FSR sensors, to determine gait-related parameters. A fixed camera, mounted on a mobile stand, is employed to record a video of an individual walking on a treadmill. The camera captures the movement of markers positioned on the subject's shoulder, hip, knee and ankle joints. The video subsequently goes through video processing techniques to compute joint angles. This is achieved by applying fundamental image processing steps to each frame of the video. The joint angles of normal and abnormal subjects are computed and compared during different gait phases.

Likewise, MPU and FSR sensors are employed to measure the movement and rotation of the lower extremities, along with measuring the distribution of force in different regions of the foot. The data acquired following the implementation and testing of the system offers a comprehensive understanding of motion-related parameters and the application of force across distinct foot regions for both normal and abnormal subjects. The analysis of this obtained data using MS-Excel forms the basis for further gait assessment and evaluation in different fields of rehabilitation and therapies for various clinical conditions.

The system’s performance can be enhanced in the future by implementing the following improvements:

1. Improved wire management and comfort

Addressing the challenge of wires is pivotal for enhancing user comfort and overall experience. An optimized approach to wire management can involve:

a. Tangle-resistant cables: Integrating tangle-resistant cables can minimize the inconvenience caused by tangled wires, allowing assessors to focus on the analysis rather than dealing with wires.

b. Strategic wire placement: Thoughtful placement of wires can prevent them from getting in the way of movement. Securing wires along the natural body contours and pathways can reduce interference and discomfort.

c. Wireless solutions: Embracing wireless technology not only streamlines the gait analysis experience but also eliminates the risk of system disruptions due to wire-related issues.

2. Specialized data analysis software

Transitioning from manual data analysis to a dedicated software tool tailored for gait analysis can elevate the analytical process. Such software can offer:

a. Automated data processing: The software can automate the processing of raw data collected from sensors, reducing the time and effort required for data preparation.

b. Advanced visualization: Visualizations like graphs and 3D models can provide a clearer representation of gait parameters, making patterns and anomalies easier to identify.

3. Stride length and symmetry analysis

Expanding the gait analysis by incorporating an additional marker on the opposite ankle can yield insightful benefits:

a. Dynamic stride analysis: Measurement of stride length can provide a dynamic understanding of how each leg contributes to the overall walking pattern, helping pinpoint any imbalances or inefficiencies.

b. Treatment tailoring: Precise data on stride length can guide the development of personalized treatment plans, focusing on optimizing each leg's contribution to gait and ensuring a more balanced and efficient walking pattern.

4. 3D Scanning

3D scanning with FSR sensors offers an innovative pathway for personalized shoe sole design. By using 3D scans to create custom-fit soles and embedding FSR sensors, the advantages include:

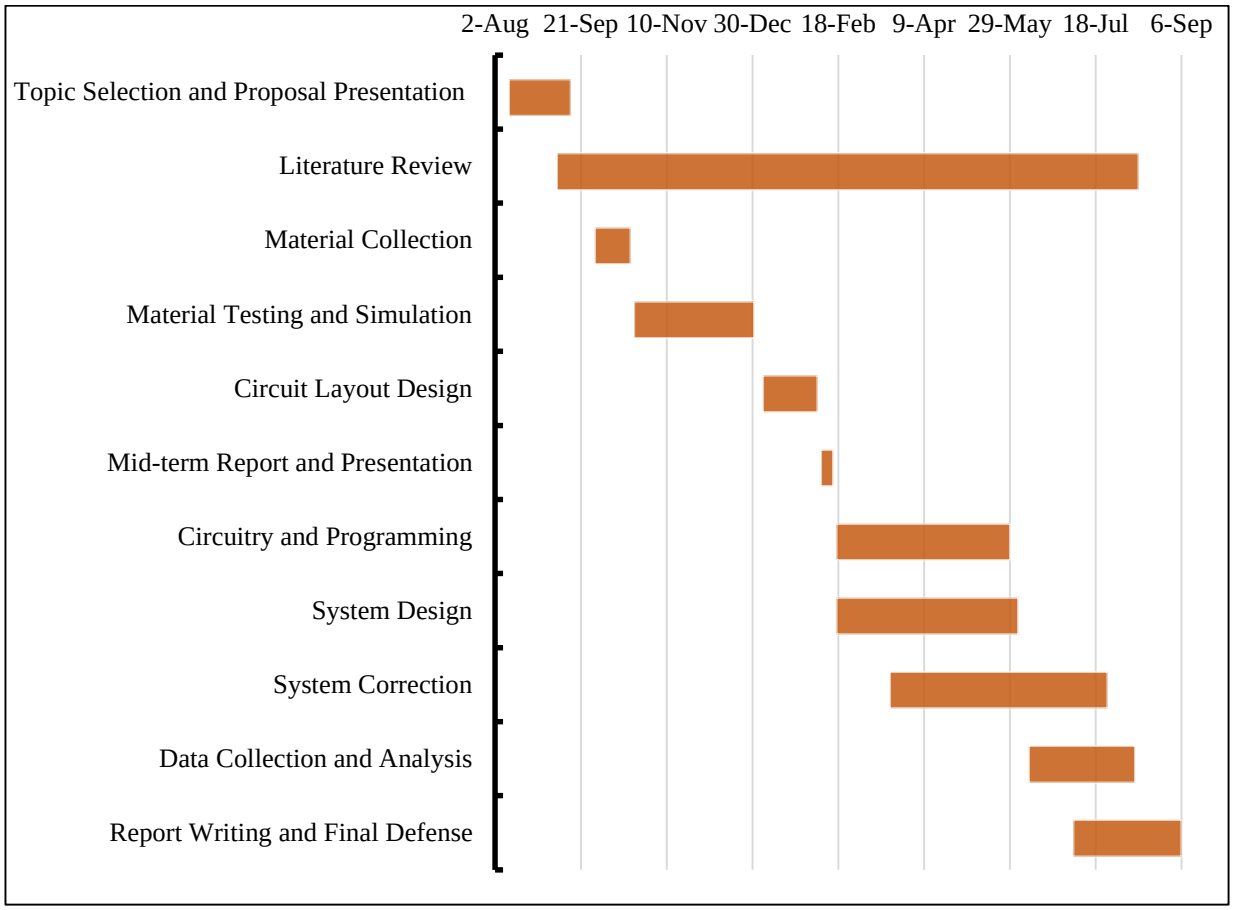
a. Custom comfort: 3D scans customize shoe soles to individual foot contours for unparalleled comfort and support.

b. Accurate force analysis: FSR sensors measure forces during walking, aiding in pinpointing pressure points and gait anomalies.

Total Cost

S.N.	Components	Quantity	Rate (Rs.)	Amount (Rs.)
1.	FSR sensors	16	650	10,400
2.	MPU-6050 sensors	10	500	5,000
3.	Arduino Mega 2560	2	2,000	4,000
4.	Bluetooth module HC-05	2	800	1,600
5.	Multiplexer TCA9548A	2	300	600
6.	Active markers	10	1,200	12,000
7.	Treadmill Repair and servicing	-	-	2,500
8.	Ribbon wire	10	100	1,000
9.	Velcro Fastener	6	50	300
10.	Resistor set	1	100	100
11.	Mobile stand	1	1,000	1,000
12.	Matrix board	8	50	400
13.	Soldering set	1	800	800
14.	Black curtain	1	1,000	1,000
15.	Bag	1	1,000	1,000
16.	Sandal	1	1,500	1,500
17.	Miscellaneous	-	-	2,000
	Total			45,200

Gantt Chart


2-Aug
21-Sep
10-Nov
30-Dec
18-Feb
9-Apr
29-May
18-Jul
6-Sep
Topic Selection and Proposal Presentation
Literature Review
Material Collection
Material Testing and Simulation
Circuit Layout Design
Mid-term Report and Presentation
Circuitry and Programming
System Design
System Correction
Data Collection and Analysis
Report Writing and Final Defense

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Appendices

Appendix 1

Python code for video processing

```
import cv2
import numpy as np
import math
import matplotlib.pyplot as plt
import openpyxl

# Load video file
video_path = 'Somiya0228x.mp4'
cap = cv2.VideoCapture(video_path)
frame_num = 1
hipAngles = []
kneeAngles = []
ankleAngles = []
trunkAngles = []
thighAngles = []
legAngles = []
angD_ft = []
feetAngles = []
final_sorted_midPoints = []

# Create a new workbook and select the active worksheet:
wb = openpyxl.Workbook()
ws = wb.create_sheet("Sheet1")
# create a new sheet with a specific name
ws.title = "Angles"  # rename the sheet

# Write the headers for the columns:
headers = ['Frame','Trunk Angle', 'Thigh Angle', 'Leg Angle', 'Foot Angle', 'Hip Angle',
'Knee Angle', 'Ankle Angle']
```

```
ws.append(headers)

name = str(input("Enter name: "))
fps = int(cap.get(cv2.CAP_PROP_FPS))
display_duration = 15

# Calculate the number of frames to display
num_display_frames = int(display_duration * fps)

# Read user input to determine which case to run
user_input = input("Press 'a' for Case 1 or 'b' for Case 2: ")

for i in range(num_display_frames):
    # Read frame
    ret, frame = cap.read()
    if ret:
        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
        blur = cv2.GaussianBlur(gray, (5, 5), cv2.BORDER_DEFAULT)
        ret, thresh = cv2.threshold(blur, 230, 255, cv2.THRESH_BINARY)
        contours, hierarchies = cv2.findContours(thresh, cv2.RETR_TREE,
cv2.CHAIN_APPROX_SIMPLE)
        midPoints = []
        for cnt in contours:
            M = cv2.moments(cnt)
            if M['m00'] != 0:
                cx = int(M['m10'] / M['m00'])
                cy = int(M['m01'] / M['m00'])
                cv2.drawContours(frame, [cnt], -1, (0, 255, 0), 2)
                cv2.circle(frame, (cx, cy), 2, (0, 0, 255), -1)
                midPoints.append((cx, cy))
        if user_input == 'a':
            # Case1: sort midpoints by y-axis and keep last two
            sorted_midPoints_y = sorted(midPoints, key=lambda x: x[1])
            last_TWO = sorted_midPoints_y[-2:]
```

```
    sorted_midPoints_x = sorted(last_TWO, key=lambda x: x[0])
    final_sorted_midPoints = sorted_midPoints_y[:-2] + sorted_midPoints_x
elif user_input == 'b':
    # Case2: sort midpoints by y-axis and keep all but last two
    sorted_midPoints_y = sorted(midPoints, key=lambda x: x[1])
    final_sorted_midPoints = sorted_midPoints_y[:-2] + sorted_midPoints_y[-2:]
# Calculate slopes
if len(final_sorted_midPoints) == 5:
    #print(f"In frame {frame_num} the final midpoints are: ",final_sorted_midPoints)
    slopes = []
    for i in range(4):
        x1, y1 = final_sorted_midPoints[i]
        x2, y2 = final_sorted_midPoints[i + 1]
        if x2 - x1 == 0:
            slope = np.inf
        else:
            slope = (y2 - y1) / (x2 - x1)
        slopes.append(slope)

# Draw lines between midpoints
    for j in range(4):
        cv2.line(frame, final_sorted_midPoints[j], final_sorted_midPoints[j+1], (0, 0,
       255), 2)

# Display frame number
    cv2.putText(frame, f"Frame: {frame_num}", (50, 50),
    cv2.FONT_HERSHEY_SIMPLEX, 1, (255, 255, 255), 2, cv2.LINE_AA)
    m1, m2, m3, m4 = slopes[0], slopes[1], slopes[2], slopes[3]

    angR_tr = math.atan(m1)
    angR_th = math.atan(m2)
    angR_lg = math.atan(m3)
    angR_ft = math.atan(m4)
```

```
if m1 < 0:
    angD_tr = -round(math.degrees(angR_tr), 2)
else:
    angD_tr = 180-round(math.degrees(angR_tr),2)
if m2<0:
    angD_th = -round(math.degrees(angR_th),2)
else:
    angD_th = 180-round(math.degrees(angR_th),2)
if m3<0:
    angD_lg = -round(math.degrees(angR_lg),2)
else:
    angD_lg = 180-round(math.degrees(angR_lg),2)
if user_input == 'a':
    if m4<0:
        angD_ft = 180-round(math.degrees(angR_ft),2)
    else:
        angD_ft = 180-round(math.degrees(angR_ft),2)
elif user_input =='b':
    if m4<0:
        angD_ft = -round(math.degrees(angR_ft),2)
    else:
        angD_ft = 180-round(math.degrees(angR_ft),2)

hipAngle = round((angD_th - angD_tr),2)
kneeAngle = round((angD_th - angD_lg),2)
ankleAngle = round((angD_ft - angD_lg - 90),2)   # type: ignore

trunkAngles.append(angD_tr)
thighAngles.append(angD_th)
legAngles.append(angD_lg)
feetAngles.append(angD_ft)
hipAngles.append(hipAngle)
kneeAngles.append(kneeAngle)
ankleAngles.append(ankleAngle)
```

```
            cv2.putText(frame, "Trunk Angle: {:.2f}".format(angD_tr), (30, 70),
            cv2.FONT_HERSHEY_COMPLEX_SMALL, 0.8, (0, 255, 255), 2)
            cv2.putText(frame, "Thigh Angle: {:.2f}".format(angD_th), (30, 90),
            cv2.FONT_HERSHEY_COMPLEX_SMALL, 0.8, (0, 255, 255), 2)
            cv2.putText(frame, "Leg Angle: {:.2f}".format(angD_lg), (30, 110),
            cv2.FONT_HERSHEY_COMPLEX_SMALL, 0.8, (0, 255, 255), 2)
            cv2.putText(frame, "Feet Angle: {:.2f}".format(angD_ft), (30, 130),
            cv2.FONT_HERSHEY_COMPLEX_SMALL, 0.8, (0, 255, 255), 2)
    else:
        break
    frame_num += 1
    cv2.imshow('Frame', frame)
    cv2.waitKey(0)

# Release video objects
cap.release()
cv2.destroyAllWindows()

data = [range(1,450),trunkAngles, thighAngles, legAngles, feetAngles, hipAngles,
kneeAngles, ankleAngles]
for row in zip(*data):
    ws.append(row)
# Save the workbook:
wb.save('angles.xlsx')

fig, (ax1, ax2) = plt.subplots(2, 1)
ax1.plot(hipAngles, label="Hip Angle")
ax1.plot(kneeAngles, label="Knee Angle")
ax1.plot(ankleAngles, label="Ankle Angle")
ax1.set_xlabel('Frame')
ax1.set_ylabel('Joint Angle (degrees)')
ax1.set_title('Lower Body Joint Angles')
ax1.legend(loc='upper right')
```

```
ax2.plot(trunkAngles, label="Trunk Angles")
ax2.plot(thighAngles, label="Thigh Angle")
ax2.plot(legAngles, label="Leg Angle")
ax2.plot(feetAngles, label="Feet Angle")
ax2.set_xlabel('Frame')
ax2.set_ylabel('Joint Angle (degrees)')
ax2.set_title('Segment Angles')
ax2.legend(loc='upper right')

fig.suptitle('Joint Angle Measurement of '+ name + ' using case ' + user_input ,
fontsize=16)
plt.subplots_adjust(hspace=0.5, bottom=0.1, top=0.9)
plt.show()
```

Arduino code for MPU sensors

```
#include <Wire.h>
#include <MPU6050.h>
#include <SoftwareSerial.h>

#define NUM_SENSORS 6
SoftwareSerial BTSerial(19, 18); // RX, TX pins of HC-05 Bluetooth module

MPU6050 mpu[NUM_SENSORS];
int16_t accelRawX[NUM_SENSORS], accelRawY[NUM_SENSORS],
accelRawZ[NUM_SENSORS], gyroRawX[NUM_SENSORS],
gyroRawY[NUM_SENSORS], gyroRawZ[NUM_SENSORS];
float accelX[NUM_SENSORS], accelY[NUM_SENSORS], accelZ[NUM_SENSORS],
gyroX[NUM_SENSORS], gyroY[NUM_SENSORS], gyroZ[NUM_SENSORS];

const int multiplexerAddress = 0x70; // I2C address of the TCA9548A multiplexer
void enableChannel(int channel);

void setup() {
  Serial.begin(9600);
  BTSerial.begin(9600);
  Wire.begin();
  Wire.setClock(400000); // Set I2C speed to 400 kHz

  // Initialize and configure the multiplexer
  Wire.beginTransmission(multiplexerAddress);
  Wire.write(0x3F); // Enable all channels (0b00111111)
  Wire.endTransmission();

  // Perform calibration
  for (int i = 0; i < NUM_SENSORS; i++) {
    enableChannel(i);
    mpu[i].initialize();
    mpu[i].setFullScaleAccelRange(0);
```

```
    delay(50);

    mpu[i].getAcceleration(&accelRawX[i], &accelRawY[i], &accelRawZ[i]);
  }
}

void loop() {
  unsigned long previousMillis = 0;
  const unsigned long interval = 100;

  for (int i = 0; i < NUM_SENSORS; i++) {
    enableChannel(i);
    mpu[i].getAcceleration(&accelRawX[i], &accelRawY[i], &accelRawZ[i]);

    accelX[i]= (accelRawX[i]) / 16384.0;
    accelY[i] = (accelRawY[i]) / 16384.0;
    accelZ[i] = (accelRawZ[i]) / 16384.0;

    // Debug output to check Bluetooth data transmission
    BTSerial.print("Sensor ");
    BTSerial.print(i + 1);
    BTSerial.print("\t");
    BTSerial.print(accelX[i]*9.81);
    BTSerial.print(" \t");
    BTSerial.print(accelY[i]*9.81);
    BTSerial.print(" \t");
    BTSerial.print(accelZ[i]*9.81);
    BTSerial.print(" \t");
  }
  BTSerial.print("\n");
unsigned long currentMillis = millis();
  unsigned long elapsedTime = currentMillis - previousMillis;
  if (elapsedTime < interval) {
    delay(interval - elapsedTime);
```

```
  }
  previousMillis = currentMillis;
  }
void enableChannel(int channel) {
  Wire.beginTransmission(multiplexerAddress);
  Wire.write(1 << channel);
  Wire.endTransmission();
}
```

Arduino code for FSR sensor

```
#include <Wire.h>
#include <SoftwareSerial.h>

#define numFSR 8
SoftwareSerial sSerial(19, 18); // RX, TX pins of HC-05 Bluetooth module

const int FSR_PINS[] = {A0, A1, A2, A3, A4, A5, A6, A7}; // Analog input pins for FSRs
const int RESISTOR_VALUE = 9800; // Resistor value in Ohm
const float GAIN = 11.892;
const float OFFSET = 24.979;
const float GRAVITY = 9.81; //in meter per second square

void setup() {
  Serial.begin(9600);
  sSerial.begin(9600);
}

void loop() {
  float Resistance[numFSR];
  float Conductance[numFSR];
  float Mass[numFSR];
  float Force[numFSR];
  float Pressure[numFSR];
  unsigned long previousMillis = 0;
  const unsigned long interval = 100;

  for (int j = 0; j < numFSR; j++) {
  int analogValue = analogRead(FSR_PINS[j]);
  Resistance[j] = ((float)RESISTOR_VALUE * (1023 - analogValue)) / analogValue;
  Conductance[j] = 1000000 / Resistance[j]; // Calculate conductance (in microS)
```

```
  // Check if Resistance is greater than 23600 and set Force to 0
  if (Resistance[j] > 23600) {
    Force[j] = 0;
  }
  else {
    Mass[j] = (Conductance[j] - OFFSET) / GAIN; // Convert conductance to mass (in kg)
    Force[j] = Mass[j] * GRAVITY;
  }
  }
  for (int j = 0; j < numFSR; j++) {
    sSerial.print("FSR ");
    sSerial.print("\t");
    sSerial.print(j + 1);
    sSerial.print("\t");
    sSerial.print(Force[j]);
    sSerial.print("\t");
  }
  sSerial.println();

  unsigned long currentMillis = millis();
  unsigned long elapsedTime = currentMillis - previousMillis;
  if (elapsedTime < interval) {
    delay(interval - elapsedTime);
  }
  previousMillis = currentMillis;
}
```

Appendix 2

Datasheet for MPU 6050

InvenSense	**MPU-6000/MPU-6050 Product Specification**	Document Number: PS-MPU-6000A-00 Revision: 3.4 Release Date: 08/19/2013

5 Features

5.1 Accelerometer Features

The triple-axis MEMS accelerometer in MPU-60X0 includes a wide range of features:

- Digital-output triple-axis accelerometer with a programmable full scale range of ±2*g*, ±4*g*, ±8*g* and ±16*g*
- Integrated 16-bit ADCs enable simultaneous sampling of accelerometers while requiring no external multiplexer
- Accelerometer normal operating current: 500µA
- Low power accelerometer mode current: 10µA at 1.25Hz, 20µA at 5Hz, 60µA at 20Hz, 110µA at 40Hz
- Orientation detection and signaling
- Tap detection
- User-programmable interrupts
- High-G interrupt
- User self-test

5.2 Additional Features

The MPU-60X0 includes the following additional features:

- 9-Axis MotionFusion by the on-chip Digital Motion Processor (DMP)
- Auxiliary master I^2C bus for reading data from external sensors (e.g., magnetometer)
- 3.9mA operating current when all 6 motion sensing axes and the DMP are enabled
- VDD supply voltage range of 2.375V-3.46V
- Flexible VLOGIC reference voltage supports multiple I^2C interface voltages (MPU-6050 only)
- Smallest and thinnest QFN package for portable devices: 4x4x0.9mm
- Minimal cross-axis sensitivity between the accelerometer and gyroscope axes
- 1024 byte FIFO buffer reduces power consumption by allowing host processor to read the data in bursts and then go into a low-power mode as the MPU collects more data
- Digital-output temperature sensor
- User-programmable digital filters for gyroscope, accelerometer, and temp sensor
- 10,000 *g* shock tolerant
- 400kHz Fast Mode I^2C for communicating with all registers
- 1MHz SPI serial interface for communicating with all registers (MPU-6000 only)
- 20MHz SPI serial interface for reading sensor and interrupt registers (MPU-6000 only)
- MEMS structure hermetically sealed and bonded at wafer level
- RoHS and Green compliant

6 Electrical Characteristics

6.1 Accelerometer Specifications

VDD = 2.375V-3.46V, VLOGIC (MPU-6050 only) = 1.8V±5% or VDD, T_A = 25°C

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS	NOTES
ACCELEROMETER SENSITIVITY						
Full-Scale Range	AFS_SEL=0		±2		*g*	
	AFS_SEL=1		±4		*g*	
	AFS_SEL=2		±8		*g*	
	AFS_SEL=3		±16		*g*	
ADC Word Length	Output in two's complement format		16		bits	
Sensitivity Scale Factor	AFS_SEL=0		16,384		LSB/*g*	
	AFS_SEL=1		8,192		LSB/*g*	
	AFS_SEL=2		4,096		LSB/*g*	
	AFS_SEL=3		2,048		LSB/*g*	
Initial Calibration Tolerance			±3		%	
Sensitivity Change vs. Temperature	AFS_SEL=0, -40°C to +85°C		±0.02		%/°C	
Nonlinearity	Best Fit Straight Line		0.5		%	
Cross-Axis Sensitivity			±2		%	
ZERO-G OUTPUT						
Initial Calibration Tolerance	X and Y axes		±50		mg	1
	Z axis		±80		mg	
Zero-G Level Change vs. Temperature	X and Y axes, 0°C to +70°C		±35			
	Z axis, 0°C to +70°C		±60		mg	
SELF TEST RESPONSE						
Relative	Change from factory trim	-14		14	%	2
NOISE PERFORMANCE						
Power Spectral Density	@10Hz, AFS_SEL=0 & ODR=1kHz		400		*μg*/ √ Hz	
LOW PASS FILTER RESPONSE						
	Programmable Range	5		260	Hz	
OUTPUT DATA RATE						
	Programmable Range	4		1,000	Hz	
INTELLIGENCE FUNCTION INCREMENT			32		mg/LSB	

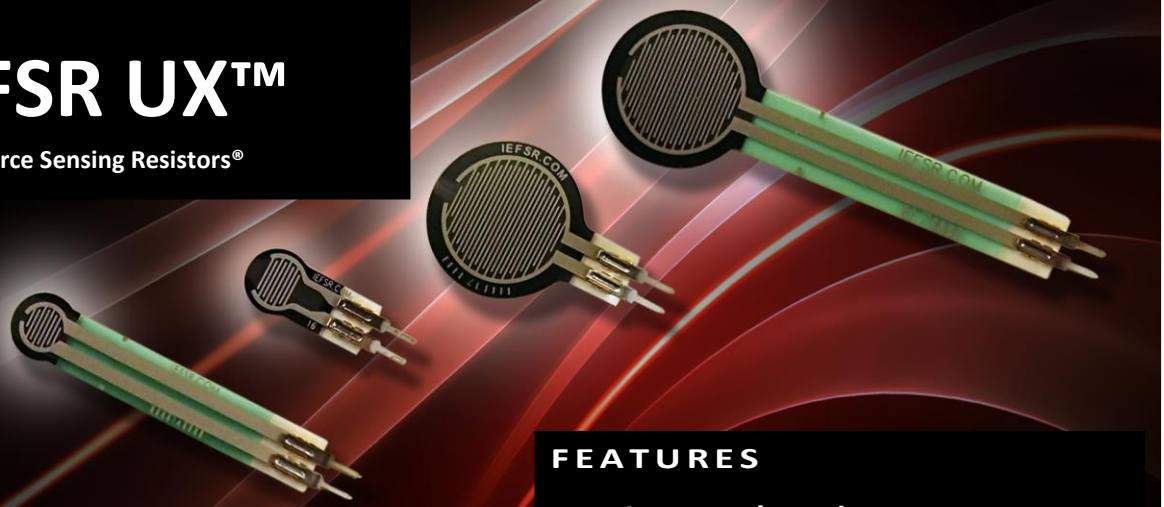
Datasheet for FSR UX 402

INTERLINK ELECTRONICS | ie SENSORS

DATASHEET

FSR UX™

Force Sensing Resistors®



Global human machine interface (HMI) leader, Interlink Electronics-the originator of the Force Sensing Resistor* (FSR)-expands its signature Line of sensors with the FSR UX.

The improved design of the FSR UX offers a greatly expanded dynamic sensing range that extends from 0.5N to 150N, providing enhanced precision and broadening the scope of the FSR UX's potential applications.

The FSR UX 400 series comprises a full range of models in various sizes, shapes, and configurations, all of which can be utilized for custom sensing solutions.

*Force Sensing Resistors are robust polymer thick film (PTF) devices that exhibit a decrease in resistance when increased force is applied to the surface of the sensor.

FEATURES

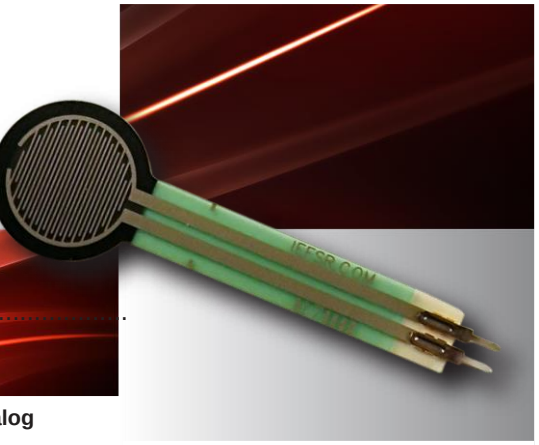
- Improved sensing range Extends from 0.5N to 150N
- Ultrathin form factor
- Able to withstand extreme Temperatures and conditions
- Tested to 10 million actuations
- Simple and easy to integrate
- Customizable to a wide range of applications
- Cost effective

APPLICATIONS

- Industrial
- Robotics
- Medical
- Automotive
- Computing
- IoT
- Wearable Technology

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FSR UX™ Force Sensing Resistors®

Device Characteristics

Sensor Resistance Output ..... **Analog**

Actuation Force..... **0.5N (min)**

Force Sensing Range ..... **0.5N – 150N**

Force Repeatability Single Part.....**+/-2%**

Force Repeatability Part to Part .....**+/-5% (Single Batch)**

No-Load Resistance.....**>10 MΩ**

Hysteresis .....**+5% (RF+ – RF-)/RF+**

Long Term Drift @ 5kg, 10 days..... **< 4% log10(time)**

Operating Temperature Performance

Cold: -25°C after 1 hour ..... **+1% resistance change**

Hot: +85°C after 1 hour ..... **+3% resistance change**

Storage Temperature Performance

Cold: -40°C after 120 hours..... **+1% resistance change**

Hot: +85°C after 120 hours ..... **+3% resistance change**

Tap Durability

Tested to 10 Million actuations @1kg, 4Hz..... **-10% resistance change**

Thermal Shock @85°C to -40°C, 20 Cycles..... **+2% resistance change**

Standing Load Durability

2.5kg for 24 hours ..... **-3% resistance change**

UL .....**All materials UL grade 94 V-1 or better**

RoHS..... **Compliant**

NOTES

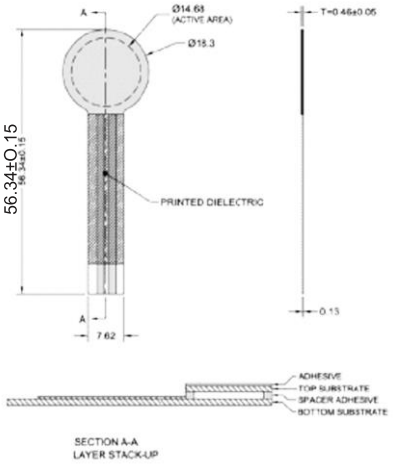
1. **Specifications are derived from average measurements taken at 5,000 grams and are given as (one standard deviation/mean), unless otherwise noted. Measurement is dependent on actuation interface, mechanics, and measurement electronics.**
2. **The test was performed using FSR UX 402 design.**
3. **Additional information is available upon request**

Sensor Mechanical Data:

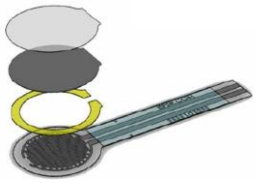
Active Area: Ø14.68mm

Nominal Thickness: 0.46mm

Switch Travel: 0.15mm



Exploded View

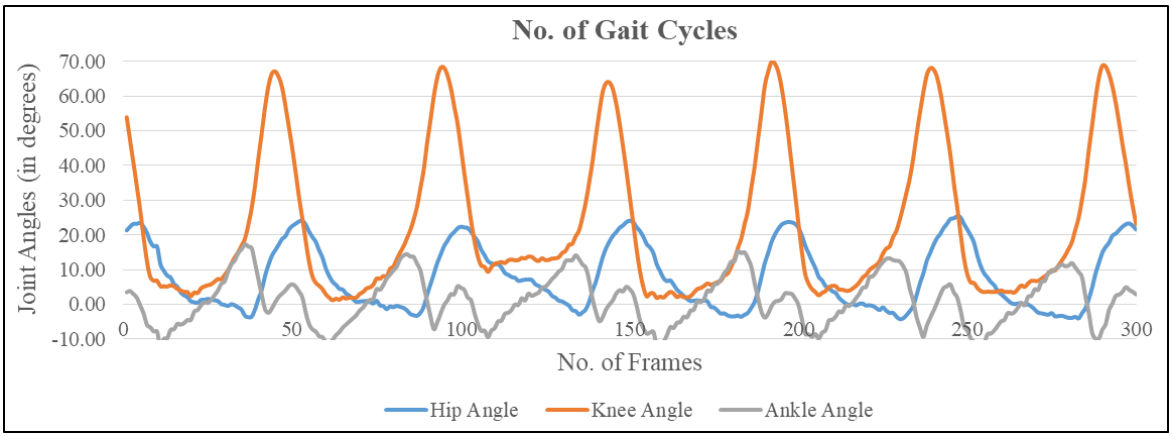


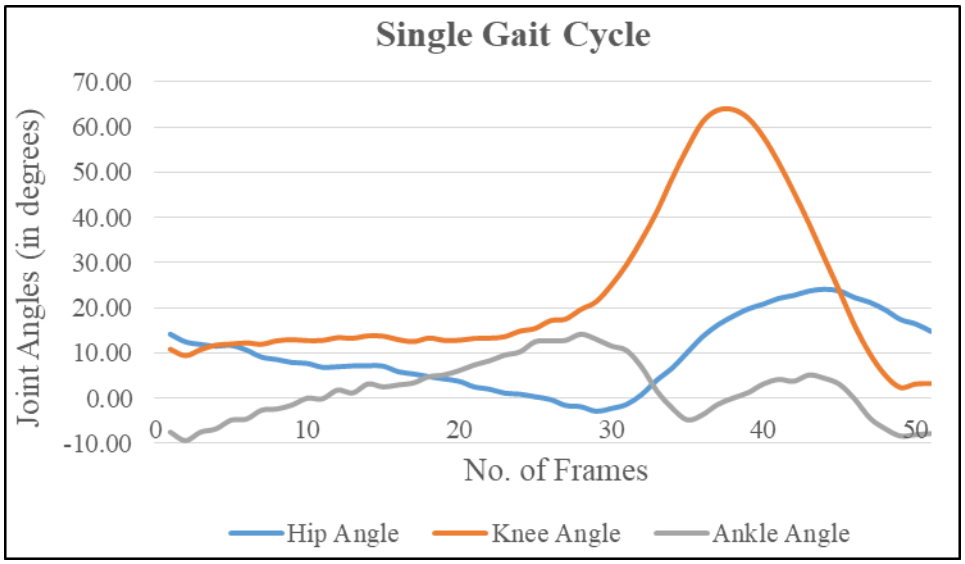
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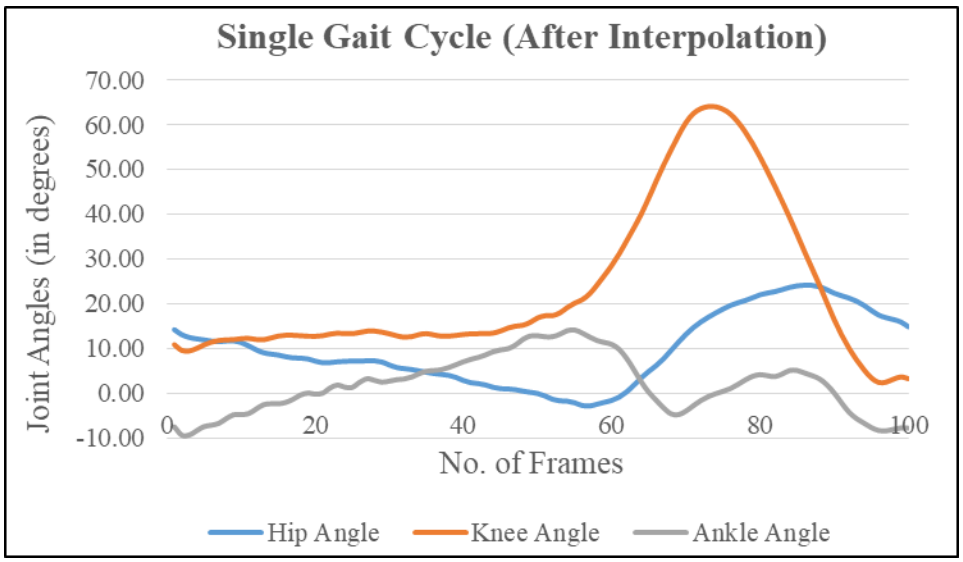
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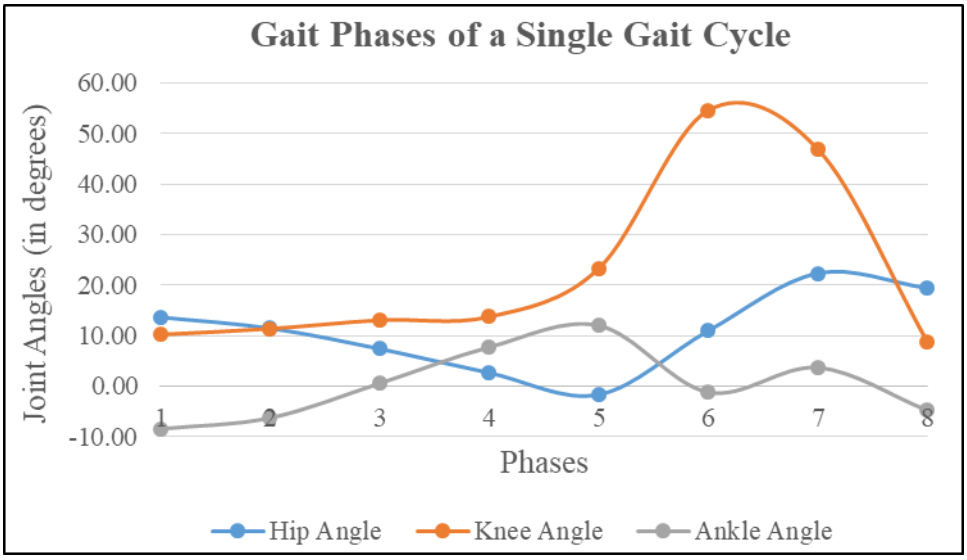
Appendix 3

Data analysis of the joint angles to evaluate one gait cycle









Numbers	Gait Phases
1	Initial contact
2	Loading response
3	Mid stance
4	Terminal stance
5	Pre swing
6	Initial swing
7	Mid swing
8	Terminal swing

Appendix 4

